

SESSION 3



Arithmetic I

Learning Objectives

1. Ratio, Proportion, Variation
2. Percentage

SESSION 03 – Arithmetic I

by PANKAJ BURHANPURKAR



SESSION 03 – Arithmetic I



Ratios

- Ratio is used for COMPARISON
- Ratio gives the relation between two or more quantities of similar type.
- $a : b$ is called as a ratio b
- $a : b = \frac{a}{b}$

Antecedent Consequent

Rules –

1. You can consider ratio as actual value by multiplying a variable into ratios.

Short-Cut

2. Units in ratio should be same

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Proper and Improper Fractions

10 : 16 Brown-haired girls to all girls

6 : 10 Blonde-haired girls to brown-haired girls

6 : 16 Blonde-haired girls to all girls

proper
fractions

10 : 6 Brown-haired girls to blonde-haired girls

16 : 10 All girls to brown-haired girls

16 : 6 All girls to blonde-haired girls

improper
fractions



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Let ratio of amount with A and B is 7 : 8, then what can be said about the amount with A and B.

Note: If only ratio is given then one cannot find the exact values.

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A: B = 3 : 4 and $A + B = 203$. Find the values of A and B.

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The salaries of A, B and C are in the ratio 5:3:2. If the increments of 20%,10%, and 20% are allowed in their salaries, then what will be the new ratio of their salaries?

Answer Options– 22 : 11 : 9, 22 : 10 : 8, 20 : 11 : 8, 20 : 10 : 9

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If $A : B : C : D = 2 : 5 : 7 : 10$, find the value of C if $A + B + C + D = 576$

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A sum of money is to be distributed among P, Q, R, and S in the ratio 2:5:6:7, If S gets Rs.500 more than R . What is P's share? **Answer Options** – 500, 700, 1000, 1200

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In a class ratio of Boys and Girls is $x : y$, then what fraction of class are boys?

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$A : B = 3 : 4$ and $B : C = 8 : 11$ then find $A : B : C$?

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The sum of the three numbers is 62. If the ratio of the first to second is 3:2 and that of the second to the third is 5:3, then the second number is

Answer Options – 20, 22, 23, 24

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If $2A = 3B = 4C$ then find $A : B : C$?

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A Zoo has 66 fishes, some white and rest orange. Which of the following could be the ratio of white to orange fishes in the zoo? **Answer Options:** 1 : 7, 2 : 9, 3 : 7, 2 : 5

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\$ 600, is distributed among A, B, C such that A gets half as much as what B and C gets together. B gets two- third as much as what C and A gets together. Find share of A?

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A person distributed “N” number (where N belongs to positive integer) of chocolates among 4 children in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{5} : \frac{1}{8}$, then

Quantity A

Quantity B

Total number of chocolates

134

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5 years ago, ratio of ages of Father and Son was 6:1 and 6 years hence ratio of their ages will be 7:2. Find present age of Son

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40 litres of solution contains water and milk in the ratio of 4: 1. Find values for following.

(A) If 6 liter solution is removed, then ratio of water and milk now.

(B) If 10 liter solution is removed and replaced by water, Find ratio of water and milk now.

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The ratio of income of Ram and Shyam is 8:11 and their expenditures are in the ratio 7:10, if each of them saves 5000, what are their income and expenditure?

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In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there? **Answer Options:** 50, 100, 150, 200

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X,Y, Z started a new venture by investing 20,000 each. After 3 months, Mr. X left the business and after 6 months Mr. Y also left the business, But Mr. Z stayed for whole year. After a year, total profit generated in the business was \$3500. Find each partner's share.

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Thanks!

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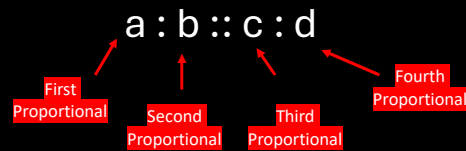
Ratio and Proportions



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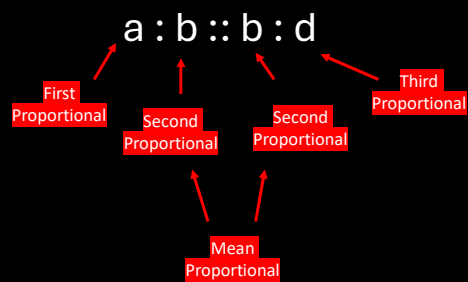
Ratio and Proportions

- $a : b :: c : d$... is called as a ratio b is proportionate to c : d
- $a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d}$



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Mean Proportionals



$$\text{Mean Proportional} = \sqrt{\text{First Proportional} \times \text{Third Proportional}}$$

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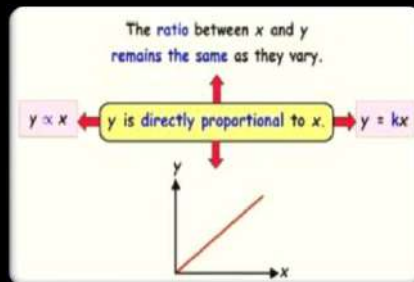
Types of proportions

Name in Latin	Theorem
Alterando	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{c} = \frac{b}{d}$
Invertendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$
Componendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{b} = \frac{c+d}{d}$
Dividendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a-b}{b} = \frac{c-d}{d}$
Componendo et Dividendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$
Addendo	$\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{g}{h} \Rightarrow \frac{a+c+e+g}{b+d+f+h}$

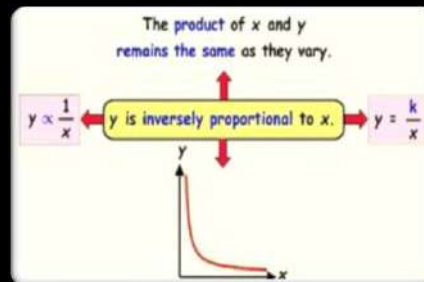
Duplicate Proportion, Triplicate Proportion,
Sub-duplicate Proportion, Sub-triplicate Proportion

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Variation, Types of Variations



Direct Variation



Inverse Variation

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If radius of a circle becomes double then Circumference and area of the circle becomes how many times?

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Cost of Diamond is directly proportional to square of its weight. A 40 gms Diamond costs \$16000. It breaks into 3 parts in ratio 1:2:5. Find reduction in cost due to breakage.

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When distance is constant speed and time are inversely proportional. When speed becomes $\frac{5}{4}$ times then final time is how many times of initial time?

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Telephone bill of Mr. Mittal is partly fixed and partly varies linearly with the number of outgoing calls. If the number of outgoing calls is 200, bill generates for \$600, But for 400 calls, it costs \$800. Find costs for Mr. Mittal if he makes 800 outgoing calls?

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The value of a silver coin varies directly as the square of its diameter, when thickness is constant and varies directly as its thickness when diameter remains constant. Two silver coins have the diameters in the ratio 4:3. Find the ratio of the thickness if the value of the first coin is four times the value of the second coin. **Answer Options:** 3:4, 4:3, 16:3, 9:4, 1:3

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At constant temperature, pressure of a definite mass of gas is inversely proportional to the volume. If the pressure is reduced by 20%, what will be the change in the volume?
Answer Options: 16.66% decrease, 25% increase, 25% decrease, 16.66% increase, None of these

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You have a rope of some length, but $\frac{2}{3}$ of it is cut off and thrown away. $\frac{1}{4}$ of the remaining rope is cut off and thrown away. What proportion of the original rope remains?
Answer Options: $\frac{1}{4}$, $\frac{1}{10}$, $\frac{1}{6}$, $\frac{1}{3}$

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60 litres of diesel is required to travel 600 km using a 800 cc engine. If the volume of diesel required to cover a distance varies directly as the capacity of the engine, then how many litres of diesel is required to travel 800 km using 1200 cc engine?
Answer Options: 80 litres, 90 litres, 120 litres, 170 litres

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Rs.432 is divided among three workers A, B and C such that 8 times A's share is equal to 12 times B's share which is equal to 6 times C's share. How much did A get?

Answer Options: Rs.192, Rs.133, Rs.144, Rs.128

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Two cogged wheels of which one has 32 cogs and other has 54 cogs, work into each other. If the latter turns 80 times in three quarters of a minute, how often does the former turn in 8 seconds? **Answer Options:** 48, 135, 24, None of these

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The monthly incomes of A and B are in the ratio 4 : 5, their expenses are in the ratio 5 : 6. If 'A' saves Rs.25 per month and 'B' saves Rs.50 per month, what are their respective incomes?
Answer Options: Rs.400 and Rs.500; Rs.240 and Rs.300, Rs.320 and Rs.400; Rs.440 and Rs.550

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The proportion of milk and water in 3 samples is 2 : 1, 3 : 2 and 5 : 3. A mixture comprising equal quantities of all 3 samples is made. The proportion of milk and water in the mixture is
Answer Options: 2 : 1, 5 : 1, 99 : 61, 227 : 133

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A group of workers can do a piece of work in 24 days. However, as 7 of them were absent it took 30 days to complete the work. How many people actually worked on the job to complete it? **Answer Options:** 35, 30, 28, 42

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A, B and C play cricket. The ratio of the runs scored by A and B is 3 : 2. The ratio of the runs scored by B and C is 3 : 2. Together, they score 342 runs. How many runs did A make?

Answer Options: 162, 108, 72, None of these

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The monthly salaries of two persons are in the ratio of 4 : 7. If each receives an increase of Rs.25 in the salary, the ratio is altered to 3 : 5. Find their respective salaries.

Answer Options: 120 and 210, 80 and 140, 180 and 300, 200 and 350

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The ratio of marks obtained by Vinod and Basu is 6 : 5. If the combined average of their percentage is 68.75 and their sum of the marks is 275, find the total marks for which exam was conducted. **Answer Options:** 150, 200, 400, None of these

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A, B and C enter into a partnership by investing Rs.3600, Rs.4400 and Rs.2800. A is a working partner and gets a fourth of the profit for his services and the remaining profit is divided among the three in the rate of their investments. What is the amount of profit that B gets if A gets a total of Rs. 8000? **Answer Options:** 4888.88, 9333.33, 4000, 3666.66

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The marks scored by a student in three subjects are in the ratio of 4 : 5 : 6. If the candidate scored an overall aggregate of 60% of the sum of the maximum marks and the maximum marks in all three subjects is the same, in how many subjects did he score more than 60%? **Answer Options:** 1, 2, 3, None of the subjects

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Thanks!

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SESSION 1 – Arithmetic I

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SESSION 1 – Arithmetic III



What is PERCENT?

- As its name suggests PER CENT means PER (for EVERY) and CENT (HUNDRED)
- Percent is to convert some quantity on the scale of 100
- In other words **PERCENT** means to divide something into **HUNDRED EQUAL PARTS**
- **$a : b :: c : 100$**

SESSION 1 – Arithmetic III

Why do we use PERCENT?

- Percentages are used for COMPARISON
- For example, if
Student A gets 350 marks out 500 and
Student B gets 320 marks out 400. Who is better?

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Base, Basic Formula and the meaning of symbol of percent (%)

- To convert any number (Fraction or Integer) in percent value divide it by base and multiply by 100
- Percent Value = $N \times \frac{100}{Base}$
- To convert from Percent to Fraction you have to divide the percent by 100
- Symbol of '%' means $\frac{1}{100}$

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Percent vs Fraction

$$\begin{aligned}
 100\% &= 100/100 = 1 \\
 50\% &= 50/100 = \frac{1}{2} \\
 33.33\% &= 33.33/100 = \frac{1}{3} \\
 25\% &= 25/100 = \frac{1}{4} \\
 20\% &= 20/100 = \frac{1}{5} \\
 16.66\% &= 16.66/100 = \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 14.28\% &= 14.28/100 = \frac{1}{7} \\
 12.5\% &= 12.5/100 = \frac{1}{8} \\
 11.11\% &= 11.11/100 = \frac{1}{9} \\
 10\% &= 10/100 = \frac{1}{10} \\
 9.09\% &= 9.09/100 = \frac{1}{11} \\
 8.33\% &= 8.33/100 = \frac{1}{12}
 \end{aligned}$$

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1%, 5%, 10% and 50% method

- | | |
|-------------------|--------------------|
| 1. 11% of 15500 = | Find 1% and 10% of |
| 2. 13% of 1700 = | |
| 3. 17% of 5000 = | 5000 |
| 4. 19% of 9990 = | |
| 5. 23% of 7000 = | |
| 6. 29% of 15000 = | 1225 |
| 7. 31% of 36000 = | |
| 8. 37% of 240 = | |
| 9. 41% of 8900 = | |
| 10. 43% of 5700 = | |

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Some simple problems

1. 30 is what percent of 240?
2. 40 is what percent of 200?
3. 200 is what percent of 1600?
4. 900 is what percent of 2700?
5. 7000 is what percent of 4200?

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Multiplying Factors

If Rohan's salary is Rs. 5000 and he gets 25% appraisal in salary what is his new salary

Rama had purchased a car for Rs. 20,000 last year. Car depreciate at the rate of 20%. What is its present price?

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Multiplying Factors

Write the multiplying factor for

80% increase

73% decrease

25% increase

16% decrease

30% increase

50% decrease

22% increase

29% decrease

200% increase

37% decrease

150% increase

94% decrease

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SESSION 1 – Arithmetic III

50 is what percent of 80?

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What is 20% of 450?

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20% of what number is 450?

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What is the result when 100 is increased by 20%?

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The number increased by 20% becoming 100. Find the number.

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A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?

Answer Options: 45%, 45.45%, 54.54%, 55%

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Two students appeared at an examination. One of them secured 9 marks more than the other and his marks were 56% of the sum of their marks. The marks obtained by them are:
Answer Options: A. 39, 30, B. 41, 32 C. 42, 33 D. 43, 34

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A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had:
Answer Options: 588 Apples, 600 Apples, 672 Apples, 700 Apples

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In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is same as of the number of students of 8 years of age which is 48. What is the total number of students in the school?

Answer Options: 72, 80, 120, 150, 100

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A student multiplied a number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation? **Answer Options:** 34%, 44%, 54%, 64%

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Base and Base Change

If Ram's salary is 20% higher than Shyam's salary then Shyam's salary is what percent less than Ram's salary

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Base and Base Change

An alloy consists of 80% Bronze and rest Brass. Bronze and Brass are also alloys with Bronze having 60% Copper and the rest Zinc whereas Brass has Copper and Zinc in equal proportions. What percentage of the alloy of Bronze and Brass is Zinc?

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Appreciation

A city's population increase at the rate of 10% per year. If the present population is 24,200 then,

- I. What was its population 2 years back
- II. What will be its population 2 years later

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Depreciation

A bike's present market price is Rs. 81,000 it depreciates at the rate of 10% per year

- I. What was its market prices two years back
- II. What will be its market price two year later

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Successive Replacement case

From a 40-litre pure milk solution, 4 litres of the solution is taken and replaced with 4 litres of water. Again from this mixture, 4 litres of mixture is taken and replaced with water. This operation is done two more time. What is the ratio of milk to water in the mixture after the fourth operation?

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Successive Percentage Change

What does it mean.... Is it FREE?

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Successive Percentage Change

If Ram's salary is first increased by 20% and then decreased by 20%, then what is the effective change in the salary?

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Successive Percentage Change

If length of rectangle is increased by 25% and breadth is decreased by 25%, what will be the effect on area?

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Successive Percentage Change

A man sold two flats for Rs. 1,00,000 each. On one he gains 16% while on the other he loses 16%. What is his percentage gain or losses in whole transaction?

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Price Increase Problems

I. If prices of a commodity increased by 20% by what percent should a person reduce its consumption so that expenditure on the commodity remain unchanged?

II. If prices of a commodity decrease by 8.33% by what percentage can the consumption should increase (quantity) so that the expenditure remain unchanged?

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The Bridge Method		
PERCENT World	FRACTION World	ACTUAL World
100%	1	\$ 10,000
50%	$\frac{1}{2}$	\$ 5,000
25%	$\frac{1}{4}$	\$ 2,500

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In a poll, a total of 1,20,000 votes were polled of which 10% were invalid. A candidate received 60% of the valid votes. Find the number of votes polled to the candidate.

In a poll, 10% of the total votes were invalid. A candidate received 60% of the valid votes and won the election by the majority of 18,000 votes. Find the total number of votes polled (There were two candidates in the election)

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If the volume of the cube is increased by 72.8%, determine the percentage increase, in the side of the cube. **Answer Options:** 10%; 20%; 50%; 72.8%

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Thanks!

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Class Practice Exercise 09

Ratio, Proportion, Variation

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. If $a : b = 5 : 9$ and $b : c = 4 : 7$, find $a : b : c$
 (1) $36 : 63 : 20$ (2) $36 : 20 : 63$
 (3) $63 : 36 : 20$ (4) $20 : 36 : 63$
2. In a school laboratory, a student notices that the time taken by a pendulum to complete one swing is directly proportional to the square root of its length. With a 9 cm long pendulum, the time was 12 seconds. What should be the length of the pendulum if the student wants the swing time to be only 4 seconds?
 (1) 1.6 cm (2) 3.6 cm
 (3) 1 cm (4) 4 cm

3. A, B, C, and D purchase a gift worth Rs.60. A pays $\frac{1}{2}$ of what others are paying, B pays $\frac{1}{3}$ rd of what others are paying and C pays $\frac{1}{4}$ th of what others are paying.

Quantity A: Twice the amount paid by D

Quantity B: Half the amount paid by C

- (1) if quantity in column A is greater;
 (2) if quantity in column B is greater;
 (3) if quantities in both the columns are equal
 (4) if the relationship cannot be determined from the information given.
4. A person distributes his pens among four friends A, B, C, D in ratio of $11 : 11 : 34 : 56$. What is the minimum number of pens that the person should have?
 (1) 175 (2) 145
 (3) 112 (4) 165
5. The fourth proportion to 4, 9 and 12
 (1) 27 (2) 36
 (3) 3 (4) $\frac{48}{9}$
6. A wizard mixes Dragon Salt and Moon Dust to prepare a magical explosive powder. Dragon Salt costs three times as much as Moon Dust. He sells the powder at ₹2160 per 10 g, earning a 20% profit. If the ingredients are mixed in the ratio $2 : 3$ (Dragon Salt : Moon Dust), find the cost price of Dragon Salt per gram.
 (1) Rs. 120/gm (2) Rs. 210/gm
 (3) Rs. 300/gm (4) None of these
7. Third proportional to 16 and 36
 (1) 81 (2) 27
 (3) $\frac{9}{4}$ (4) Can't determine
8. The mean proportional between 0.08 and 0.18.
 (1) 0.12 (2) 0.0144
 (3) 0.13 (4) 0.108

9. The cost of insurance on a house is directly proportional to the area of the house. Alan has a 2,500 square-foot house and pays a monthly insurance premium of $\$66\frac{2}{3}$. Bella has a 3,750 square-foot house insured with the same company.

Quantity A: Yearly premium paid by Bella

Quantity B: 1200

- (1) if quantity in column A is greater;
 (2) if quantity in column B is greater;
 (3) if quantities in both the columns are equal
 (4) if the relationship cannot be determined from the information given.
10. The distance of the horizon at sea varies as the square root of the height of the eye above sea-level. When the distance is 14.4 km, the height of the eye is 18 meters. Find, in kilometers, the distance when the height of the eye is 8 meters.
 (1) 12 km (2) 9.6 km
 (3) 7.2 km (4) 4.8 km
11. In a car factory, the daily manufacturing cost of a batch of 500 brake pads is partly fixed and partly variable. At standard production, each brake pad costs ₹5. When production rises by 40%, the cost per pad drops to 90% of the original. (Variable cost is directly proportional to output.)
Quantity A: Variable manufacturing cost per brake pad.
Quantity B: 3
 (1) if quantity in column A is greater;
 (2) if quantity in column B is greater;
 (3) if quantities in both the columns are equal
 (4) if the relationship cannot be determined from the information given.
12. A space mission training center has costs that are partly fixed and partly proportional to the number of astronauts in training. When there are 25 astronauts, the average cost per person is ₹70. When the number rises to 50, the average reduces to ₹60. What will be the average cost per astronaut if 100 trainees join?
 (1) 54 (2) 50
 (3) 55 (4) 56
13. A certain amount was to be distributed among A, B and C in the ratio $2 : 3 : 4$, but was erroneously distributed in the ratio $7 : 2 : 5$. As a result of this, B received Rs. 40 less.
Quantity A: Amount of distribution
Quantity B: 420
 (1) if quantity in column A is greater;
 (2) if quantity in column B is greater;
 (3) if quantities in both the columns are equal
 (4) if the relationship cannot be determined from the information given.

14. If $x : y = 3 : 4$, find $4x + 5y : 5x - 2y$
 (1) 32 : 7 (2) 31 : 8
 (3) 28 : 11 (4) 16 : 7
15. If three times P, four times Q and five times R bear the ratio of 6 : 7 : 8 with each other, what would be the ratio that P, Q and R bear with each other?
 (1) 168 : 192 : 210 (2) 6 : 7 : 8
 (3) 120 : 105 : 96 (4) 18 : 28 : 40
16. The King of Aryavrat leaves behind ₹60,000 crores of royal treasure. The treasure is to be shared between princes and princesses in the ratio 7 : 5. Among the princes, the share of Prince Veer to Prince Arjun is 4 : 3. How much does Prince Arjun receive?
 (1) 25,000 Crores (2) 15,000 Crores
 (3) 22,000 Crores (4) 12,000 Crores
17. Divide Rs. 672 in the ratio of 5 : 3.
 (1) ₹420 and ₹252 (2) ₹400 and ₹272
 (3) ₹448 and ₹224 (4) ₹360 and ₹312
18. The oscillation period of a pendulum is inversely proportional to the square root of its length. At a time $O = 5$ seconds and length of the pendulum = 64 cm Find the length if $O = 4$ seconds.
 (1) 100 cm (2) 75 cm
 (3) 60 cm (4) 80 cm
19. Divide Rs. 1162 among A, B, C in the ratio 35 : 28 : 20.
 (1) Rs. 448 (2) Rs. 462
 (3) Rs. 490 (4) Rs. 504
20. The cost of diamond varies directly to the square of its weight. At a time, the cost and weight 50 mgs. If the diamond breaks such that weights of its pieces are in the ratio 3:2. What will be the reduction in the cost of diamond?
 (1) Rs. 4800 (2) Rs. 2400
 (3) Rs. 2600 (4) Rs. 4000
21. A bag contains 50 p, 25 p and 10 p coins in the ratio 5 : 9 : 4, amounting to Rs. 206. Find the number of coins of each type.
 (1) 200, 360, 160 (2) 180, 324, 144
 (3) 220, 396, 176 (4) 250, 450, 200
22. A, B and C invested their capital in the ratio 2 : 3 : 5 and the time period of their investment being in the ratio 4 : 5 : 6. In what ratio will the profit be divided?
 (1) 8 : 12 : 25 (2) 8 : 15 : 30
 (3) 4 : 9 : 18 (4) 16 : 15 : 12
23. A mixture contains alcohol and water in the ratio 4:3. If 5 litre of water is added to the mixture, the ratio becomes 4:5. Find the quantity of alcohol in the given mixture.
 (1) 8 L (2) 9 L
 (3) 10 L (4) 12.5 L
24. A's present age to B's present age is 8:7. 27 years ago their ages were in the ratio 5:4. Find their present ages.
 (1) 56, 49 (2) 32, 28
 (3) 72, 63 (4) 80, 70
25. If $A : B = 5 : 7$ and $B : C = 6 : 11$, then $A : B : C$ is
 (1) 30 : 42 : 77 (2) 35 : 49 : 42
 (3) 55 : 77 : 66 (4) None of these
26. 6 parts of milk are mixed with 1 part of water, 5 liters of water are added to the mixture and it is then found that milk is five times as much as water. How many liters of water were there in the mixture originally?
 (1) 16 (2) 20
 (3) 25 (4) None
27. If $A : B = 3 : 4$ and $B : C = 8 : 9$, then $A : C$ is
 (1) 1 : 2 (2) 2 : 3
 (3) 3 : 2 (4) 1 : 3
28. Divide 27 into two parts such that 5 times the first part and 11 times the second part are together equal to 195.
 (1) 15, 12 (2) 17, 10
 (3) 16, 11 (4) None
29. If $A : B = 8 : 15$, $B : C = 5 : 8$ and $C : D = 4 : 5$, then $A : D$ is equal to
 (1) 8 : 15 (2) 15 : 4
 (3) 2 : 7 (4) 4 : 15
30. A dog takes 4 leaps for every 5 leaps of a hare but 3 leaps of dog is equal to 4 leaps of hare. Compare their speeds.
 (1) 22 : 24 (2) 19 : 20
 (3) 17:18 (4) 16:15
31. If $A : B : C = 2 : 3 : 4$, then $\frac{A}{B} \cdot \frac{B}{C} \cdot \frac{C}{A}$ is equal to
 (1) 4 : 9 : 16 (2) 8 : 9 : 12
 (3) 8 : 9 : 16 (4) 8 : 9 : 24
32. The mean proportional between 0.04 and 25 is
 (1) 2 (2) 4
 (3) 3 (4) 1
33. If $A : B = \frac{1}{2} : \frac{3}{8}$, $B : C = \frac{1}{3} : \frac{5}{9}$ and $C : D = \frac{5}{6} : \frac{3}{4}$, then the ratio $A : B : C : D$ is –
 (1) 16 : 24 : 30 : 35 (2) 18 : 24 : 30 : 35
 (3) 16 : 22 : 30 : 35 (4) 16 : 24 : 15 : 35
34. Profit of Rs. 84 is divided among A and B in the ratio $\frac{1}{3} : \frac{1}{4}$. What will be the share of each?
 (1) 30, 54 (2) 37, 37
 (3) 48, 36 (4) 58, 26
35. If $A : B = 2 : 3$, $B : C = 4 : 5$ and $C : D = 6 : 7$, then $A : B : C : D$ is
 (1) 14 : 16 : 18 : 10 (2) 16 : 24 : 30 : 35
 (3) 16 : 18 : 9 : 10 (4) 18 : 16 : 10 : 9

36. In the mixture of 50 liters, milk and water are in the ratio 3:2. How much water should be added to make the ratio equal?
 (1) 5 (2) 20
 (3) 15 (4) 10
37. If $2A = 3B = 4C$, then $A : B : C$ is
 (1) 20 : 15 : 2 (2) 6 : 4 : 3
 (3) 4 : 3 : 2 (4) 2 : 3 : 4
38. A, B and C enter into a partnership with investments in the ratio 5:7:8. If at the end of the year A's profit is Rs. 42360 Find the total profit.
 (1) 169440 (2) 210000
 (3) 183000 (4) 196700
39. If $\frac{P}{3} = \frac{Q}{4} = \frac{R}{5}$, then $P : Q : R$ is
 (1) 3 : 4 : 5 (2) 20 : 15 : 2
 (3) 4 : 3 : 5 (4) 5 : 4 : 3
40. A charity organization distributes a fund among four welfare projects. The first project receives half of the total fund. The second project receives one-fourth. The third project receives one-fifth. And the fourth project receives ₹9000. Find the total amount of the fund.
 (1) Rs. 220000 (2) Rs. 100000
 (3) Rs. 90000 (4) Rs. 180000
41. If $2A = 3B$ and $4B = 5C$, then $A : C$ is
 (1) 3 : 4 (2) 15 : 8
 (3) 8 : 15 (4) 4 : 3
42. $A : B = 3 : 4$ and $B : C = 5 : 6$ find $A : B : C$
 (1) 3:2:7 (2) 15:20:24
 (3) 2:3:4 (4) None of these
43. Given $a : b = 6 : 5$, $a : c = 15 : 11$ find the ratio $a : b : c$
 (1) 21 : 5 : 11 (2) 6 : 5 : 11
 (3) 90 : 75 : 66 (4) 30 : 25 : 22
44. The ratio $4^{35} : 2^5$ is same as
 (1) $2^{35} : 1$ (2) $4^{65} : 1$
 (3) $7^{65} : 5$ (4) $2^{65} : 1$
45. The ages of A and B are in the ratio 6:5 and the sum of their ages is 44 years. Find the ratio of their ages after 8 years.
 (1) 3 : 4 (2) 1 : 2
 (3) 7 : 6 (4) 8 : 7
46. $\frac{1}{5} : \frac{1}{x} = \frac{1}{x} : \frac{1}{125}$, then the value of x is
 (1) 15 (2) 20
 (3) 25 (4) 35
47. The ratio of Fighter Ships to Cargo Freighters to Exploration Cruisers in a newly assembled galactic fleet is 10:2:3. If the fleet contains 120 Fighter Ships, what is the total number of Exploration Cruisers?
 (1) 40 (2) 25
 (3) 36 (4) None of these
48. If $0.75 : x :: 5 : 8$, then x is equal to
 (1) 1.12 (2) 1.2
 (3) 1.25 (4) 1.30
49. The time it takes for a rescue mission to complete varies inversely as the number of rescue helicopters deployed. When 2 helicopters are deployed, the mission takes 300 minutes (5 hours). If 6 helicopters are deployed, how many minutes will the mission take?
 (1) 200 people (2) 400 people
 (3) 100 people (4) 150 people
50. The ratio of 1 hr 20 mins and 50 minutes, is
 (1) 1 : 2 (2) 15 : 5
 (3) 8 : 5 (4) 4 : 3
51. A comic book enthusiast, Alex, collects two types of comic books: Super-Hero Comics and Sci-Fi Comics. The number of Super-Hero Comics to Sci-Fi Comics is in the ratio 5 : 2. If Alex has 450 more Super-Hero Comics than Sci-Fi Comics, what is the total number of Sci-Fi Comics in the collection?
 (1) 300 (2) 120
 (3) 150 (4) 200
52. If the total number of students in a class is 40 and 25 of them are girls, find the total number of boys to the number of girls.
 (1) 5 : 3 (2) 3 : 5
 (3) 3 : 8 (4) 5 : 8
53. An architect is designing a courtyard using two different types of square tiles - large Marble Tiles and smaller Granite Tiles. The ratio of the side length of the Marble Tile (M) to the side length of the Granite Tile (G) is 4 : 3. If the sum of the areas of the two tiles is 225 square inches, what is the difference between the side length of the Marble Tile and the side length of the Granite Tile?
 (1) 5 (2) 6
 (3) 3 (4) 4
54. Divide Rs. 72 in the ratio 2 : 3 : 4
 (1) 18, 27, 36 (2) 14, 21, 28
 (3) 32, 24, 16 (4) 16, 24, 32

55. A jewel thief's secret stash contains 500 total gems, consisting only of Emeralds (green) and Rubies (red). The ratio of Emeralds to Rubies is 3:2. By how many is the number of Emeralds greater than the number of Rubies?
- (1) 200 (2) 300
(3) 100 (4) None of these
56. If $a : b = 3 : 4$ and $b : c = 5 : 6$, then the ratio $a : b : c$
- (1) 3 : 4 : 6 (2) 18 : 24 : 20
(3) 15 : 20 : 24 (4) 9 : 12 : 15
57. An archaeologist discovers two fragments of pottery. The length of the first fragment (The 'Elder' Piece) to the length of the second fragment (The 'Younger' Piece) is in the ratio of 7 : 3. If the length difference between the two fragments is 28 centimeters, what is the length of the Elder Piece?
- (1) 56 cm (2) 42 cm
(3) 49 cm (4) 35 cm
58. If $a : b = 2 : 5$, $b : c = 3 : 5$ and $c : d = 5 : 4$, then the ratio $a : b : c : d$.
- (1) 2 : 5 : 3 : 4 (2) 6 : 15 : 25 : 20
(3) 6 : 15 : 75 : 60 (4) 2 : 15 : 25 : 4
59. In 6 dozen of Kiwis, some kiwis are rotten. Which of the following cannot be ratio of number of kiwis that are rotten to that are good. (Choose one or more)
- (1) 1 : 8 (2) 7 : 1
(3) 3 : 7 (4) 5 : 2
(5) 13 : 5 (6) 1 : 1
60. The ratio of the income and expenditure of Amit and Sumit are 5 : 8 and 11 : 10. If Amit and Sumit save Rs. 7900 and Rs. 21,000 respectively, find their incomes. Two integers whose sum is 64, cannot be in ratio
- (1) 1:7 (2) 3:5
(3) 5:11 (4) 1:2
61. Which of the following are correct: (Choose one or more)
- (1) $2^{(n-1)} + 2^{(n+1)} = 320$ if $n = 8$.
(2) Value of $0.04^{-1.5}$ is 125
(3) If a, b, c are distinct natural numbers and $a > 1$, then : $a^{b^c} > (a^b)^c$
(D) If 22 pounds of cake consists of flour, cream, sugar, and Choco chips in ratio 5 : 3 : 1 : 2 . Then quantity of cream in the cake will be 3 pounds.
62. A pirate captain finds 40 pounds of gold ingots. He decides to divide the treasure among his two most trusted crew members, Salty and Pegleg, in the ratio of their years of service, 5:3. What is the weight of the gold ingots that Salty (the first person in the ratio) receives?
- (1) 15 (2) 25
(3) 8 (4) 20
63. An ancient alchemist is working with a total of 94 grams of two rare metals, Electrum and Orichalcum. He finds that if he takes one-fifth of the Electrum's weight and compares it to one-eighth of the Orichalcum's weight, the resulting ratio is 3 : 4. What is the weight, in grams, of the Electrum?
- (1) 40 (2) 30
(3) 64 (4) 32
64. A sum of money is to be distributed among A, B, C, D in the proportions of 5 : 2 : 4 : 3. If C gets 4000 more than D, what is B's share?
- (1) 16,000 (2) 8,000
(3) 12,000 (4) 20,000
65. What is the difference (in dollars) between the largest and the second smallest part when a sum of \$676 is divided into four parts having the ratio of 5 : 7 : 6 : 8?
- (1) 26 (2) 52
(3) 104 (4) 130
66. The three sides of the triangle are in ratio 3 : 6 : 7 and the perimeter is 112 cm. Determine the largest side.
- (1) 21 (2) 42
(3) 49 (4) 56
67. A space scientist is studying two newly discovered planets, Planet Alpha (A) and Planet Beta (B). The masses of Planet Alpha and Planet Beta are in the ratio of 3:4. The scientist discovers a critical relationship: eight times the square of Alpha's mass is 224 units greater than the square of Beta's mass. What are the respective masses of Planet Alpha and Planet Beta?
- (1) 9, 12 (2) 12, 9
(3) 12, 16 (4) 6, 8
68. The ratio of the number of boys and girls in a college is 7 : 8. If the percentage increase in the number of boys and girls is 20% and 10% respectively, what will be the new ratio?
- (1) 14 : 15 (2) 77 : 80
(3) 21 : 22 (4) 22 : 21
69. A fort has provisions for 60 days. If after 15 days 500 men strengthen them and then the food lasts 40 days. How many men were there in the fort at beginning?
- (1) 3500 (2) 4000
(3) 6000 (4) None
70. The cost of a diamond is directly proportional to the square of its weight. A certain diamond broke into pieces in the ratio 1 : 2 : 3 : 4. On selling the pieces the merchant got \$700 less than the original price of the diamond. Find the original price of the diamond
- (1) 700 (2) 1000
(3) 300 (4) 7000

71. Satellite deceleration (a) is inversely proportional to t^2 . If $a = 5 \text{ ly/m}^2$ when $t=20$ minutes, find a when $t = 25$ minutes.
 (1) 5.5 ly/m^2 (2) 2.25 ly/m^2
 (3) 4.5 ly/m^2 (4) 3.2 ly/m^2
72. If a solution contains water to soap in ratio 34 : 15 and water to acid in ratio 51 : 56. Determine the ratio of water to soap to acid in the solution.
 (1) 102 : 45 : 112; (2) 45:112:102;
 (3) 112:45:102; (4) 135:42:12
73. The output ratings ratio of two sources T:L is 6:5. The sum of their actual outputs is 275 GJ. If the combined average efficiency (Actual Output / Maximum Capacity) is 68.75%, what is the maximum capacity of one source??
 (1) 150 GJ (2) 400 GJ
 (3) 200 GJ (4) 600 GJ
74. 6 parts of milk are mixed with 1 part of water, 5 liters of water are added to the mixture and it is then found that milk is five times as much as water. How many liters of water were there in the mixture originally?
 (1) 26 (2) 10
 (3) 25 (4) None
75. In the mixture of 50 liters, milk and water are in the ratio 3:2. How much water should be added to make the ratio equal?
 (1) 25 (2) 35
 (3) 40 (4) 10
76. Three partners shared the profit in a business in the ration 5 : 7: 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio fo their investments.
 (1) 20 : 49 : 64 (2) 70 : 56 : 56
 (3) 5 : 7 : 8 (4) 40 : 90 : 128
77. The current ratio of the height of an Ancient Greek Vase V_1 to the height of an Egyptian Urn U_2 is 1:2. Five centuries ago, the ratio of their heights (due to restoration/erosion) was 1:3. What will be the ratio of their heights five centuries from now?
 (1) 1 : 4 (2) 2 : 3
 (3) 3 : 5 (4) 5 : 6
78. A, B and C rent a pasture. A puts 10 oxen for 7 months. B puts 12 oxen for 5 months and C puts 15 oxen for 3 months for grazing. If the rent of the pasture is Rs. 175, find the share of C in the rent.
 (1) 80 (2) 60
 (3) 45 (4) 70
79. Divide Rs. 5130 among P, Q, R and S such that 3 times P' share = 4 times Q's share = 5 times R's share = 6 times S's share. What is the share of Q (in Rs.)
 (1) 1800 (2) 1350
 (3) 1080 (4) 900
80. In a party, males and females are in the ratio of 5:2. If there are 450 more males than the females then how many females are there in the party?
 (1) 150 females (2) 120 females
 (3) 300 females (4) 200 females

Class Practice Exercise – 10

Ratio, Proportion, Variation II

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. A vessel contains 180 L wine. 60 L is taken out of the vessel every day and an equal volume of water is put in it. What volume of wine remains in the vessel at the end of 3 days?
 (1) $160/3$ L (2) $20/3$ L
 (3) 80 L (4) 35 L
2. The sum of three consecutive even numbers is 56 more than the sum of two consecutive odd numbers, which lie between the even numbers. Find the average of the five numbers.
 (1) 43 (2) 112
 (3) 56 (4) 84
3. A solution contains 20 L of pure milk. 20% of this is taken out. Another 20% of the remaining solution is taken out and finally 20% of the solution left is taken out. The total amount of solution taken out is replaced with water. Find the ratio of milk to water in the final solution.
 (1) 1 : 5 (2) 12 : 38
 (3) 488 : 512 (4) 512 : 488
4. If $\frac{a}{b} = \frac{1}{3}, \frac{b}{c} = 2, \frac{c}{d} = \frac{1}{2}, \frac{d}{e} = 3, \frac{e}{f} = \frac{1}{4}$, then what is the value of $\frac{abc}{def}$?
 (1) $\frac{3}{8}$ (2) $\frac{27}{8}$
 (3) $\frac{3}{4}$ (4) $\frac{27}{4}$
5. The combined ages of the parents of a girl is five times as that of the girl's. After 8 years, their combined ages would be four times that of the girl. What was the combined ages of the parents when the girl was born?
 (1) 80 years (2) 64 years
 (3) 48 years (4) 72 years
6. If $\frac{a}{b} = \frac{2}{3}, \frac{b}{c} = \frac{3}{5}$ and $\frac{c}{d} = \frac{5}{6}$ then what is the value of $\frac{3a+2b+4c+d}{3d+4c}$?
 (1) $19/6$ (2) 38
 (3) 1 (4) 2
7. If $\frac{a}{b} = \frac{3}{4}, \frac{b}{c} = 5, \frac{c}{d} = \frac{4}{3}, \frac{d}{e} = 2$ and $\frac{e}{f} = \frac{1}{4}$ then $\frac{def}{abc} =$
 (1) $40/9$ (2) $400/9$
 (3) $9/40$ (4) $9/400$
8. Monthly incomes of X and Y are in the ratio 1 : 3 and their expenses are in the ratio 19 : 40. X saves Rs. 18,860 less than that Y and in total they save Rs. 36,020. Income of X and Y respectively are:
 (1) Rs. 10,480 and Rs. 31,440
 (2) Rs. 9,000 and Rs. 27,000
 (3) Rs. 14,200 and Rs. 42,600
 (4) Rs. 18,000 and Rs. 31,440
9. A beaker contains milk and water in the ratio 5 : 3. If a person removes 6 L of this mixture and replaces it with pure water, the ratio of milk to water becomes 1 : 1. What is the amount of milk present in beaker initially?
 (1) 18.75 L (2) 16.25 L
 (3) 20 L (4) 18 L
10. A container has 80 L of milk. From this container, 8 L of milk was taken out and replaced by water. The process was further repeated once. How much milk is there in the container now?
 (1) 64.8 L (2) 62.8 L
 (3) 62 L (4) 64 L
11. A rope is 64 m long. It is cut into 3 pieces. The longest piece is 3 times as long as the middle-sized piece and the shortest piece is 34 m shorter than the longest piece. Find the length of the middle-sized piece.
 (1) 14 m (2) 16 m
 (3) 18 m (4) 22 m
12. The ratio of the first-class fare and second-class fare is 3:1 and that of the number of passengers traveling by the first class and by the second class is 1:27. If Rs. 2,700 is collected as fare, then the amount collected from the first-class passengers is
 (1) Rs. 300 (2) Rs. 729
 (3) Rs. 270 (4) Rs. 96.42
13. Let a, b, c, d and e be integers such that $a = 12b = 24c$ and $4b = 15d = 249$. which of the following pairs contains a number that is definitely not an integer?
 (1) $\left(\frac{a}{18}, \frac{b}{e}\right)$ (2) $\left(\frac{b}{c}, \frac{a}{de}\right)$
 (3) $\left(\frac{b}{6}, \frac{cd}{e}\right)$ (4) $\left(\frac{a}{4b}, \frac{d}{e}\right)$
14. The cost of three books and twelve pens is Rs. 27. If the cost of two books and x pens is 18, then find the value of x.
 (1) 4 (2) 12
 (3) 8 (4) 16
15. A cylindrical container is having the top surface of a liquid at a level of $4/7$ of its full level. From this, if x litres is removed, the container is one-fourth full. Finally, 35 L is added to the container from this position, making it half full. Find the value of x and the capacity of the container respectively.
 (1) 45 L and 210 L (2) 25 L and 140 L
 (3) 45 L and 140 L (4) 40 L and 210 L
16. The ratio of the present ages of Chinta and Mani is 14 : 19. The average age of Suknaya, Chinta and Mani is 43 years. If the ratio of present age of Suknaya to that of Chinta is 5 : 7, find the present age (in years) of Mani.
 (1) 28 (2) 42
 (3) 57 (4) 56

17. If $\frac{ax^2+by^2}{ax^2-by^2} = \frac{77}{13}$ and $\frac{xy^2+ab^2}{xy^2-ab^2} = \frac{17}{7}$, what is the value of $(x^2 : b^2)$
 (1) 1 : 4 (2) 9 : 4
 (3) 2 : 3 (4) 49 : 64
18. In an organization, ratio of the number of males to that of the females is 4 : 3. If the number of males is increased by 10% and that of females by 20%. What is the new ratio of the number of males to that of females?
 (1) 11 : 9 (2) 10 : 9
 (3) 11 : 8 (4) 3 : 2
19. If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$ and $d = \frac{-b}{2}$ and $f = \frac{-2d}{5}$, then $\frac{a+2c+5e}{b+2d+5f} =$
 (1) $\frac{a+2c}{b+2d}$ (2) $\frac{3c+2e}{3d+2f}$
 (3) $\frac{2c+5e}{2d+5f}$ (4) All of the above
20. The present ages of a father and his son are in the ratio 9 : 4. The ratio of the father's age after 8 years from now to the age of the son 2 years ago is 8 : 3. Find the present age (in years) of the son.
 (1) 32 (2) 40
 (3) 16 (4) 72
21. Which of the following could be the relation between the speed(s) of a vehicle and its weight (w) given that $w = 27$ tonnes for $s = 5$ m/sec and $w = 8$ tonnes for $s = 20/9$ m/sec?
 (1) $W \propto S^{2/3}$ (2) $S \propto W^{2/3}$
 (3) $W \propto S^{1/3}$ (4) $S \propto W^{5/3}$
22. A student scored marks in the ratio 5 : 4 : 6 : 8 : 7 in five subjects having equal maximum marks. In all, he scored 50% of the maximum marks in all the five subjects taken together. In how many subjects did he score more than 55% of the maximum marks?
 (1) 1 (2) 2
 (3) 3 (4) 5
23. Three cups of ice-cream, two burgers and four soft drinks together cost Rs. 128. Two cups of ice-creams, one burger and two soft-drinks together cost Rs. 74. What is the cost of each cup of ice-cream?
 (1) Rs. 20 (2) Rs. 30
 (3) Rs. 25 (4) Can't determine
24. When three brands of juices are mixed in the ratio of 3 : 4 : 5 and 4 : 5 : 6, the cost price of the two piece mixtures comes out to be Rs. 20 per litre and Rs. 25 per litre respectively. Find the cost price of a litre of mixture of juices in which the three brands of original juices are mixed in the ratio of 6 : 7 : 8.
 (1) Rs. 30.7 (2) Rs. 30.5
 (3) Rs. 29.5 (4) Rs. 29.7
25. Birbal says to Akbar, "I am thrice as old as you were when I was as old as you are". If the sum of their present ages is 80 years, then how many years ago was Birbal twice as old as Akbar?
 (1) 12 (2) 10
 (3) 16 (4) Can't determine
26. A family consists of father, mother, son and daughter. The ratio of the weight of the father to that of the son is 3 : 2. The ratio of the weight of the son to that of the mother is 5 : 6. If the weight of the daughter is 35 kg, which is half the weight of the father, find the weight of the mother.
 (1) 42 kg (2) 56 kg
 (3) 46.6 kg (4) Can't determine
27. The value of a certain kind of precious stone is proportional to the square of its weight. One such stone is broken into four pieces, the weights of which are in the ratio 1 : 2 : 3 : 4. If a loss of Rs. 14,000 is incurred due to the breakage, find the value of the original stone.
 (1) Rs. 20,000 (2) Rs. 1,40,000
 (3) Rs. 30,000 (4) Rs. 28,000
28. One year ago, the ratio of Harsha's and Mandar's salaries was 3 : 5. The ratios of their individual salaries of last year and present year are 2 : 3 and 4 : 5 respectively. If their combined salary for the present year is Rs.86,000, find the present salary of Harsha.
 (1) Rs.38,000 (2) Rs.24,000
 (3) Rs.39,000 (4) Rs.36,000
29. If $(a+2b) : (b+2c) : (c+2a) = 5 : 6 : 7$ and $a + b + c = 18$, then find the value of a.
 (1) 6 (2) 8
 (3) 7 (4) 5
30. The numbers of students in Batch A and Batch B were in the ratio 2 : 3 in January and 5 : 8 in February. The numbers of students in Batch A and Batch B increased from February to March at rates that were twice and thrice, respectively, of rates at which they increased from January to February. If the ratio of the total number of students in these two batches in February and January was 26 : 5, then find the ratio of number of students in Batch A and Batch B in March?
 (1) 25 : 64 (2) 25 : 84
 (3) 26 : 55 (4) 35 : 92
31. If the cost of one pen, one pencil and three erasers is Rs. 140 and the cost of five pens, three pencils and an eraser is Rs. 320, then find the cost of three pens, two erasers and two pencils.
 (1) Rs. 460 (2) Rs. 230
 (3) Rs. 250 (4) Can't determine
32. Total expenses of a boarding house are partly fixed and partly varying linearly with the number of boarders. The average expense per boarder is Rs. 700 when there are 25 boarders and Rs. 600 when there are 50 boarders. What is the average expense per boarder when there are 100 boarders?
 (1) 550 (2) 580
 (3) 540 (4) 570

33. The ratio of the number of sparrows on the left and right branches of a tree is 7 : 13. If 12 sparrows shift to the left branch from the right, then the number of sparrows on the left branch is equal to that on the right branch. How many sparrows were there on the left branch, before the shift? (Assume that the branches of the tree are classified as either left branches or right branches.)
 (1) 84 (2) 14
 (3) 28 (4) 40
34. The height of Santosh is directly proportional to the square root of his age. What will be the height of Santosh at 16 years, if he is 4 ft tall at the age of 9 years?
 (1) 5 ft 4 inches (2) 5 ft 2 inches
 (3) 5 ft (4) 5 ft 3 inches
35. In a three-digit number, the number obtained when twice the tens digit is subtracted from thrice the hundreds digit, is equal to the number obtained when the units digit is subtracted from thrice the hundreds digit which in turn is equal to the number obtained when twice the tens digit is subtracted from the units digit. How many such three-digit numbers are possible?
 (1) 0 (2) 1
 (3) 2 (4) 3
36. Abhishek and Rani started a partnership business by investing Rs.1 cr and Rs.2 cr respectively in the beginning of the year. Each of them withdrew one-fourth of their investment after each quarter till 2nd quarter. At this point Susmita and Randeep joined them with Rs.50 lakh and Rs.1 cr respectively. Randeep withdrew his entire investment after 3 months. If the profit at the end of the year is Rs. 41 lakhs, the combined share of Abhishek and Rani is
 (1) Rs. 22 lakh (2) Rs. 33 lakh
 (3) Rs. 8 lakh (4) Rs.30 lakh
37. Ten years hence, the age of a man will be five years less than twice the age of his only daughter. Ten years ago, his wife's age was fifteen years more than twice the age of their daughter. Between the wife and husband who is older and by how many years?
 (1) Husband, 6 years (2) Wife, 5 years
 (3) Wife, 4 years (4) None of these
38. In a mixture of 120 L, the ratio of milk and water is 2 : 3. If 30 L of mixture is replaced by 18 L of milk and 12 L of water, then new ratio of milk to water is
 (1) 7 : 9 (2) 9 : 11
 (3) 2 : 3 (4) 11 : 9
39. If the equations $px + qy + r = 0$ and $tx + my + n = 0$; yield a unique solution set for the values of x and y , then which of the following is necessarily true?
 I. $(pm \neq ql)$ II. $(qn \neq rm)$
 III. $(rl \neq pn)$
 (1) III alone (2) I alone
 (3) I, II and III (4) II alone
40. Two liquids A and B are mixed in the ratio of 3 : 2 respectively and the mixture is sold at Rs. 11 per litre at 10% profit. If the first liquid costs Rs. 2 per litre more than the second, the cost (rupees per litre) of first liquid is
 (1) 10.8 (2) 8.8
 (3) 18.8 (4) 5
41. Which of the following series of numbers is in the descending order?
 (1) $\frac{1024}{1019}, \frac{5321}{5316}, \frac{512}{507}, \frac{7327}{7322}$ (2) $\frac{5321}{5316}, \frac{7327}{7322}, \frac{512}{507}, \frac{1024}{1019}$
 (3) $\frac{512}{507}, \frac{1024}{1019}, \frac{5321}{5316}, \frac{7327}{7322}$ (4) $\frac{7327}{7322}, \frac{5321}{5316}, \frac{1024}{1019}, \frac{512}{507}$
42. A milkman mixes 20 litres of water with 80 litres of milk. After selling one-fourth of this mixture, he adds water to replenish the quantity that he had sold. What is the current ratio of water to milk?
 (1) 2 : 3 (2) 1 : 2
 (3) 1 : 3 (4) 3 : 4
43. A piece of land is to be divided between two men in the ratio of 5 : 11. Instead, if the land is divided in the ratio 11 : 5, what fraction of the whole piece of land does the second man lose?
 (1) $\frac{3}{8}$ (2) $\frac{6}{11}$
 (3) $\frac{5}{6}$ (4) $\frac{10}{11}$
44. Gold is 18 times as heavy as water and copper is 8 times as heavy as water. The ratio in which copper to gold be mixed so that the mixture is 15 times as heavy as water, is
 (1) 3 : 2 (2) 7 : 3
 (3) 3 : 7 (4) 2 : 3
45. The numbers a , b and c are positive integers. Three times a is equal to seven times b and four times b is equal to three times c . Which of the following cannot be the value of $a + 2b + 3c$?
 (1) 1725 (2) 3050
 (3) 5710 (4) 1925
46. Two alloys of iron have different percentages of iron in them. The first of them weighs 6 kg and the second one weighs 12 kg. One piece each, of equal weight, was cut off from both the alloys. The piece from the first was mixed with the leftover part of the second alloy and the piece from the second with the leftover part of the first alloy. As a result the percentage of iron became the same in both the alloys. What was the weight of each cut out pieces?
 (1) 4 kg (2) 3.5 kg
 (3) 3 kg (4) None of these
47. The engine of a railway train can cover a distance of 30 km in 20 minutes, when no wagons are attached to it. When wagons are attached, the decrease in speed is directly proportional to the cube root of the number of wagons attached. When 8 wagons are attached. It covers a distance of 14 km in 12 minutes. What is the speed of the train (in kmph) when 64 wagons are attached to the engine?
 (1) 60 (2) 70
 (3) 50 (4) 30

48. There is an alloy (1) of silver and copper. A certain weight of this alloy is mixed with 15 kg of pure silver and melted. The new alloy (2) contains 90% of silver. If the alloy (1) is mixed with 10 kg of a 90% silver alloy, the new alloy (3) is found to contain 84% silver. Find the percentage of silver in a alloy (1).
- (1) 80% (2) 90%
(3) 75% (4) 85%
49. In a fraction, if 5 is added to the numerator and 7 to the denominator, the fraction becomes equal to $\frac{2}{3}$. If 2 is subtracted from the denominator and 1 from the numerator, the fraction becomes equal to $\frac{2}{3}$. Find the difference between the numerator and the denominator.
- (1) 2 (2) 4
(3) 3 (4) Can't determine
50. An alloy of copper and zinc contains 82% copper. After 18 kg of zinc had been added to the alloy, the copper content gets reduced to 70%. How much copper and zinc does the new alloy contain?
- (1) Copper 88 kg, zinc 35 kg
(2) Copper 86.1 kg, zinc 36.9 kg
(3) Copper 80.5 kg, zinc 42.5 kg
(4) Copper 70 kg, zinc 30 kg
51. Him and Haw, to do their MBA program at Roseiur Institute, appeared for the FIAT entrance exam, and got 7 and 24 marks respectively. Each question answered correctly gets one mark but the negative marks for each of the first five wrong attempts is different from that of the remaining wrong attempts. Him attempted 22 questions and Haw 45 questions and their strike rate is 50% and 66.66% respectively. Find the negative marks for each mistake made, beyond the first 5 mistakes.
- (1) $\frac{1}{3}$ (2) $\frac{1}{2}$
(3) $\frac{1}{5}$ (4) $\frac{1}{4}$
52. The concentrations of the milk solutions in three containers A, B and C are in an arithmetic progression, such that $CA > CB > CC$ (C indicates concentration). The volume of the containers are VA, VB and VC. Which of the following statement(s) is/are true?
- I. If $VA : VB : VC = 1 : 1 : 1$, then the volumes of milk are also in arithmetic progression.
II. If the amount of milk in the containers is in AP, it implies that the volumes of the containers are all equal.
III. There exists situations when the concentration of a combined mixture can be independent of the volume VB (i.e. not affected by VB).
- (1) Statements I and II are correct
(2) Statements I and III are correct
(3) Statements II and III are correct
(4) Only statement II is correct
53. A garrison of 2,000 men has provisions for 20 weeks at the rate of 2.5 kg per day per man. After 4 weeks, 500 more men join the garrison. For how many more weeks will the remaining provisions last at the rate of 2 kg per day per man?
- (1) 12 weeks (2) 15 weeks
(3) 16 weeks (4) 20 weeks
54. Pipe X pours a mixture of acid and water, and pipe Y pours pure water into a bucket. After 1 hour, the bucket got filled and the concentration of acid in the bucket was noted to be 8%. If pipe Y was closed after 30 minutes and pipe X continued to pour the mixture, concentration of acid in the bucket after 1 hour would have been 10%. What is the ratio of acid to water in the mixture coming out of pipe X?
- (1) 13 : 2 (2) 2 : 15
(3) 3 : 20 (4) 2 : 13
55. Two positive integers p and q are in the ratio 3 : 7. If 9 and 21 are subtracted respectively from p and q the ratio of the resulting numbers, which are also positive, taken in the same order, is again 3 : 7. What is the second least value of (p + q)?
- (1) 20 (2) 40
(3) 60 (4) 50
56. Three milkmen Ramu, Kamu and Namu had 5L, 10L and 15L of milk-water solution respectively. In the solution that Kamu and Namu had, the ratio of milk and water were 2 : 3 and 1 : 4 respectively. All of them sat together and mixed their respective solutions together. They sold the combined mixture at Rs. 20 per litre and Ramu got Rs. 2.5 for every litre of mixture sold. What was the milk-water ratio in the solution that Ramu had with him? (Assume water comes free of cost and they share the revenue earned in the ratio of milk content)
- (1) 1 : 4 (2) 2 : 3
(3) 1 : 5 (4) 3 : 5
57. Three times x added to twice y is 22 and the product of x and y is 20. If $3x + 5y + z = 43$ and x, y and z are integers, then find the average of x, y, z.
- (1) 5 (2) $\frac{5}{3}$
(3) $\frac{15}{2}$ (4) $\frac{37}{6}$
58. Two jars contain milk and water in the ratio 5 : 4 and 2 : 1 respectively. What volume should be taken out from the first jar and second jar so as to fill up a third 30 L jar with milk to water in the ratio 1 : 1?
- (1) 7.5 L (2) 15 L
(3) 22.5 L (4) Impossible situation
59. A parent buys 5 erasers for every 3 pencils bought by him. For every 24 articles of the combined lot of erasers and pencils bought by him, he buys 2 geometry boxes. If the total number of all three types of articles bought by him is 104, what is the number of pencils bought by him?
- (1) 65 (2) 36
(3) 8 (4) 60
60. Infinite 1-L containers A0, A1, A2... exist. A0 starts with 1 L alcohol; others empty. In each step, half of A0 is taken out and equally poured into A1...An (step n), and A0 is refilled with water. After infinitely many steps, what is the alcohol concentration in A1?
- (1) 62% (2) 66%
(3) 75% (4) 80%

61. The cost of three apples, two custard apples and three mangoes is Rs. 55. If thrice the cost of the apple is Rs. 4 more than that of the custard apple and the ratio of the cost of the apple to the custard apple is 3 : 5. Then find the total cost of two apples, three custard apples and four mangoes.
- (1) Rs. 69 (2) Rs. 65
(3) Rs. 68 (4) Rs. 138
62. Three vessels contain sugar solution in volumes forming an AP with common difference 5 kg. All have the same sugar percentage. The difference between this percentage and the volume of the smallest vessel is 8. The total sugar in all three vessels is 5.4 L. Find the volumes of the three vessels.
- (1) 5 kg, 10 kg and 15 kg (2) 6 kg, 11 kg and 16 kg
(3) 7 kg, 12 kg and 17 kg (4) 8 kg, 13 kg and 18 kg
63. Hariharan's expenditure is equal to Lezy Louis's salary and his savings are half of Lezy Louis savings. If Lezy Louis's salary is Rs. 25000, then which of the following is definitely true?
- (1) The sum of twice Hariharan's salary and Lezy Louis's expenditure is Rs. 75000.
(2) Hariharan's salary is not less than Rs. 25,000 but not more than Rs. 37500
(3) If Hariharan's salary is more than Rs. 37500, then Lezy Louis expenditure could be Rs. 0.
(4) Both (1) and (2)
64. Beaker A and beaker B contains methanol, ethanol and phenyl in the ratio 1 : 3 : 2 and 2 : 1 : 5 respectively. Some parts of the solutions from beaker A and beaker B are thoroughly mixed and put into another beaker C. Which of the following cannot be the ratio of methanol, phenyl and ethanol in beaker C?
- (1) 10 : 23 : 15 (2) 7 : 15 : 16
(3) 6 : 13 : 13 (4) 9 : 20 : 18
65. 5 kg of sugar costs the same as that of 2 litres of cooking oil. 8 kg of rice can be bought with the same amount as is needed to buy 3 litres of cooking oil. Two kg of dal and 5 kg of rice have the same price. If 1 litre of cooking oil costs Rs. 40, then what is the difference between the price of 1 kg of dal and kg of sugar?
- (1) 321.50 (2) Rs. 22.50
(3) Rs. 12.50 (4) Rs. 25
66. There are two alloys of copper and aluminium in the ratio 7 : 2 and 7 : 11 respectively. If equal quantities of the two alloys are melted to form a third alloy, then find the ratio of copper to aluminium in newly formed alloy.
- (1) 3 : 4 (2) 5 : 3
(3) 7 : 3 (4) 7 : 5
67. Four brothers — Ram, Lakshman, Bharath and Shathrugna — bought a car for Rs. 2, 40, 000. Ram paid half of the sum of the amounts paid by his three brothers. Shathrugna paid one-fifth of the total amount paid by the other three brothers. The amount paid by Bharath is five-nineteenth of the total amount paid by his three brothers. How much did Lakshman pay?
- (1) Rs. 50,000 (2) Rs. 40,000
(3) Rs. 70,000 (4) Rs. 60,000
68. An empty container is filled with pure alcohol. The alcohol is slowly allowed to run out and when the container is $\frac{1}{4}$ empty, it is replaced with water. Next when the container is half empty it is again filled with water. Finally, when it is $\frac{3}{4}$ empty, it is again filled with water. Find the percentage of alcohol in the container now.
- (1) 37.5% (2) 11.75%
(3) 9.375% (4) 14.375%
69. Consider the equations $px + 3y = 12$ and $2x + qy = r$ where p, q and r are distinct single digit natural numbers. Find the number of triplets (p, q, r) for which the given pair of equations have no solution.
- (1) 4 (2) 26
(3) 28 (4) 34
70. After the addition of 35 liters of water to a can of diluted milk, the concentration of milk in the can becomes 30%. Now, further 40 liters of water is added to the can and the concentration of milk in the can gets reduced by 10 percentage points. How many more liters of water must be added to the can now such that the concentration of milk in the can becomes 8%?
- (1) 120 lit (2) 150 lit
(3) 180 lit (4) 200 lit

Class Practice Exercise – 11

Ratio, Proportion, Variation III

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

- The price of a particular brand of a refrigerator in 2010 was Rs. 9600 whereas that in 2011 was Rs. 10800 with the price increasing by a fixed amount each year. Express the price of the refrigerator (in Rs.) in terms of T, the number of years from the base year 2004
 (1) 2000T (2) 4000 + 200T
 (3) 1200T (4) 1200 (T + 2)
- If $\frac{5m-4n}{5m+4n} = \frac{1}{4}$ and $3m + 2n = 24$, then m and n respectively are
 (1) $4, \frac{16}{3}$ (2) $\frac{13}{3}, \frac{11}{2}$
 (3) $\frac{11}{3}, \frac{13}{2}$ (4) $\frac{16}{3}, 4$
- A can contains a solution of two liquids A and B in the ratio of 7 : 5. When 9 L of solution is drawn off from the solution and the can is filled with B, the proportion of A and B becomes 7 : 9. How many litres of liquid A was there in the can initially?
 (1) 14 L (2) 21 L
 (3) 28 L (4) 35 L
- The salaries of A, B and C are in the ratio 1 : 2 : 5. If the increments of 20%, 15% and 10% respectively are allowed in their salaries, then what will be the new respective ratio of their salaries?
 (1) 12 : 23 : 55 (2) 11 : 23 : 55
 (3) 23 : 11 : 55 (4) 12 : 55 : 23
- In a mixture of 105 L, the ratio of milk and water is 8 : 7. If some amount of mixture is removed in which 14 L was water and 5 L of water is added afterwards into the remaining mixture, then the new ratio of milk to water is
 (1) 19 : 20 (2) 4 : 5
 (3) 1 : 1 (4) 2 : 3
- One year ago, the ratio between Harsha's and Mandar's salaries was 3 : 5. The ratios of their individual salaries of last year and present year are 2 : 3 and 4 : 5 respectively. If their combined salary for the present year is Rs.86,000, find the present salary of Harsha.
 (1) Rs.38,000 (2) Rs.24,000
 (3) Rs.39,000 (4) Rs.36,000
- In what ratio should 30% Phenyl Alcohol should be mixed with 60% of Phenyl Alcohol solution so that the resultant solution has 50% Phenyl Alcohol in it?
 (1) 1 : 2 (2) 2 : 1
 (3) 1 : 3 (4) 3 : 1
- Rs.74,000 is distributed among x, y and z in the ratio $\frac{1}{4} : \frac{1}{5} : \frac{1}{6}$. The share of y is
 (1) Rs.36,000 (2) Rs.24,000
 (3) Rs.27,000 (4) Rs.30,000

- A solution of 20 L of wine and water contains 80% wine. How much water (in L) must be added to this solution in order to make it a 40% wine solution?
 (1) 20 L (2) 16 L
 (3) 30 L (4) 12 L
- A person has a total of Rs.210 in one-rupee, 50-paisa and 25-paisa coins. The number of the three types of coins are in the ratio 2.5 : 3 : 4 respectively. How many one-rupee coins has the person got?
 (1) 89 (2) 103
 (3) 92 (4) 105
- A, B and C invested their money in the ratio 3 : 6 : 7. If the total amount invested by them was Rs. 80,000 and the profit earned was 40% of the amount invested. Then what was the sum of the shares of B and C in the profit?
 (1) Rs. 18,000 (2) Rs. 26,500
 (3) Rs. 26,000 (4) Rs. 24,000
- The height of Santosh is proportional to the square root of his age. What will be the height of Santosh at 16 years, if he is 4 ft tall at the age of 9 years?
 (1) 5 ft 4 inches (2) 5 ft 2 inches
 (3) 5 ft (4) 5 ft 3 inches
- Ajay started a firm with a capital of Rs. 28,000. After 5 months, Boman joined him and invested Rs. 40,000 in the firm. Chirag was also added as a new partner with an individual investment of Rs. 56,000 after 7 months of commencement. If at the end of the year, the profit of the firm is Rs. 32,000, what is the share of Boman?
 (1) Rs. 12,000 (2) Rs. 8,000
 (3) Rs. 14,000 (4) Rs. 10,000
- The pressure of a gas is directly proportional to its temperature and inversely proportional to its volume. The initial pressure, volume and temperature are 150 pascals, 20 cu. m and 100° kelvin respectively. Find the final temperature if the final volume and pressure of the same gas are 8 cu. m and 150 pascals respectively under the same conditions.
 (1) 80° kelvin (2) 50° kelvin
 (3) 40° kelvin (4) 45° kelvin
- In a business partnership among A, B, C and D, the profit is shared as follows:

$$\frac{A's\ share}{B's\ share} = \frac{B's\ share}{C's\ share} = \frac{C's\ share}{A's\ share}$$
 If the total profit is Rs. 4,00,000, the share of C is
 (1) Rs. 1,12,500 (2) Rs. 1,33,333
 (3) Rs. 90,000 (4) Rs. 2,70,000
- Y is the sum of three quantities which are directly proportional to X, X², X³ respectively, and when X = 1, Y = 4, when X = 2, Y = 8, and when X = 3, Y = 18. What is the value of Y when X = 7?
 (1) 98 (2) 128
 (3) 198 (4) 238

17. A began a business with Rs. 4,500 and was joined after wards by B with Rs. 5,400. If the profits at the end of the year were divided in the ratio 2 : 1, then B joined the business after:
- (1) 5 months (2) 4 months
(3) 6 months (4) 7 months
18. 100 million bacteria can completely decompose a garbage dump in 15 days and 60 million bacteria can do so in 30 days. If same quantity of garbage is added to the initial quantity of the dump every day, how many bacteria will be required to completely decompose the dump in 10 days?
- (1) 150 Mn. (2) 180 Mn.
(3) 140 Mn. (4) 120 Mn.
19. Three people A, B and C entered into partnership and the ratio of their investments was $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$. The ratio of the time period for which they invested their money was $1 : \frac{1}{2} : \frac{1}{4}$. The profits were shared in the direct proportion to amount invested and time. If A's share is Rs. 30,000, then find B's share.
- (1) Rs. 10,000 (2) Rs. 20,000
(3) Rs. 15,000 (4) Rs. 25,000
20. The total cost of a trip is constant. The cost per head reduces from Rs.25 to Rs.20 when 20 more people join in. Find how much it would further reduce if 25 more people join this new group.
- (1) Rs. 4 (2) Rs. 5
(3) Rs. 8 (4) Rs. 16
21. A is the average of a series of n consecutive odd numbers starting with x and B is the average of n consecutive odd numbers starting with x + 4. The value of (B – A)
- (1) is always 4 (2) depends on x
(3) depends on n (4) is always a multiple of 4
22. Three partners A, B and C invested Rs. 6,000, Rs. 7,000 and Rs. 8,000 respectively in a trading firm. Find their respective shares, if the total profit is Rs. 3,360.
- (1) Rs. 960, Rs. 1,120, Rs. 1,280
(2) Rs. 800, Rs. 900, Rs. 1,660
(3) Rs. 860, Rs. 1,000, Rs. 1,500
(4) Rs. 1,200, Rs. 1,080, Rs. 1,080
23. X, Y and Z are partners in a business with a total capital of Rs. 3,300. The profit at the end of the year is Rs. 1,500 to be divided in proportion to their capitals. If X receives Rs. 450 and Y receives Rs. 550 as their share of profits, then Z's capital is
- (1) Rs. 910 (2) Rs. 1,100
(3) Rs. 1,110 (4) None of these
24. A, B and C enter into partnership. A invests 4 times as much as B invests and B invest three-fourth of what C invests. At the end of the year, the total profit earned is Rs. 7,220. What is the share of A?
- (1) Rs. 1,520 (2) Rs. 1,140
(3) Rs. 4,450 (4) Rs. 4,560
25. A, B and C invested Rs. 2,000, Rs. 5,000 and Rs. 4,000 respectively in a business. The net profit for the year was Rs. 1,210 which was divided in proportion to investment. Then the amount of profit earned by A, B and C respectively, is
- (1) Rs. 220, Rs. 550, Rs. 420
(2) Rs. 220, Rs. 450, Rs. 440
(3) Rs. 220, Rs. 550, Rs. 440
(4) Rs. 210, Rs. 450, Rs. 440
26. If $x : y = 1 : 2$, then $(4x + 3y) : (7x - 3y)$ is
- (1) 10 : 1 (2) 1 : 5
(3) 10 : 3 (4) 1 : 3
27. A, B and C invested their money in the ratio 3 : 6 : 7. If the total amount invested by them was Rs. 80,000 and the profit earned was 40% of the amount invested. Then what was the sum of the shares of B and C in the profit?
- (1) Rs.18,000 (2) Rs.26,000
(3) Rs.26,500 (4) Rs.24,000
28. $\frac{x}{y} = \frac{z}{w} = \frac{p}{q} = 1.25$, then what is the value of $\frac{2x+3z+4p}{2y+3w+4q}$
- (1) 1 (2) 1.25
(3) 1.5 (4) 1.75
29. A can contains a solution of two liquids A and B in the proportion of 7 : 5. When 9 L of solution is drawn out and the can is filled with B, the proportion of A and B in the final mixture becomes 7 : 9. How many litres of liquid A was there in the can initially?
- (1) 21 L (2) 31 L
(3) 80 L (4) 85 L
30. Harshal invested Rs. 3,000 for 8 months while Kaushal invested Rs. 4,000. If the final profit is divided in the ratio 2 : 1 respectively, then Kaushal invested for how much duration?
- (1) 3 months (2) 5 months
(3) 6 months (4) 8 months
31. The pressure of a gas is directly proportional to its temperature and inversely proportional to its volume. The initial pressure, volume and temperature are 150 pascals, 20 cu. m and 100° Kelvin respectively. Find the final temperature if the final volume and pressure of the same gas are 8 cu. m and 150 pascals respectively under the same conditions.
- (1) 80° Kelvin (2) 50° Kelvin
(3) 40° Kelvin (4) 45° Kelvin
32. If $28 : x :: x : 7$, then find the value of x.
- (1) 7 (2) 14
(3) 28 (4) 196
33. A is directly proportional to B. B is inversely proportional to C. C is directly proportional to the square of D. D is directly proportional to the cube root of E. Assuming all are positive integers, as A increases, E
- (1) increases (2) decreases
(3) remains constant (4) doubles

34. A sum of Rs.4,980 is to be divided among A, B and C in the ratio $\frac{1}{10} : \frac{1}{9} : \frac{1}{4}$ respectively. Find B's share.
- (1) Rs.600 (2) Rs.1,200
(3) Rs.1,800 (4) Rs.120
35. A person buys some apples and mangoes from the market. Cost price of a mango is twice that of an apple and the selling price of a mango is thrice that of an apple. By selling an apple at twice its cost price, he makes 150% profit on the whole. Find the ratio of the number of mangoes to that of apples that he bought from the market.
- (1) 3 : 5 (2) 3 : 4
(3) 1 : 2 (4) 2 : 3
36. The average age of three children is 10 years and their ages are in ratio 2 : 3 : 5. The age of the middle child is
- (1) 15 years (2) 9 years
(3) 6 years (4) 30 years
37. A man sells chocolates that come in boxes. Either full boxes or half a box of chocolates can be bought from him. A customer comes and buys half the number of boxes the seller has, plus half a box. A second customer comes and buys half the remaining number of boxes plus half a box. After this, the seller is left with no chocolate box. How many chocolate boxes did the seller have before the first customer came?
- (1) 2 (2) 3
(3) 4 (4) 3.5
38. The age of a man and his son is in the ratio 4 : 1. Five years hence the respective ratio of their ages will be 3:1. How old is the father at present?
- (1) 40 years (2) 60 years
(3) 75 years (4) 100 years
39. The ratios of the incomes and the expenditures of Aishwarya, Babita and Charu are 7 : 9 : 12 and 8 : 9 : 15 respectively. If Aishwarya saves one-fourth of her income, then the ratio of their savings is
- (1) 56 : 99 : 69 (2) 69 : 56 : 99
(3) 99 : 56 : 69 (4) 99 : 69 : 56
40. If $\frac{x}{y} = \frac{a}{b} = \frac{3}{2}$, then what is $\frac{x^2+a^2}{y^2+b^2}$?
- (1) $\frac{13}{4}$ (2) $\frac{9}{4}$
(3) $\frac{13}{9}$ (4) $\frac{12}{9}$
41. The value of a diamond is directly proportional to the square of its weight. If a diamond weighing 4 kg breaks into 2 pieces, its total value decreases by 37.5%. Weights of the two pieces are:
- (1) 2 kg, 2 kg (2) 2.5 kg, 1.5 kg
(3) 3 kg, 1 kg (4) None of these
42. The total profit of Rs.3,600 is to be distributed amongst A, B and C such that ratio of shares of A and B is 5 : 4 and that of B and C is 8 : 9. What is the share of C?
- (1) Rs.1,200 (2) Rs.1,350
(3) Rs.1,500 (4) Rs.1,800
43. The sum of four numbers is 253. The ratio of the first number to the second one is 2 : 3. The ratio of the second number to the third one is 5 : 6. The ratio of the third number to the fourth one is 8 : 9. What is the average of the second and the third number?
- (1) 65 (2) 66
(3) 67 (4) 68
44. Father's age was five times his son's age 5 years ago and will be three times the son's age after 2 years. What is the ratio of the present age of father to that his son?
- (1) 10 : 3 (2) 7 : 2
(3) 4 : 1 (4) 9 : 2
45. When two students of a class with strength of 40 students were expelled, the average age of the class did not change. If one of the students was 14 years old and average of the class was 18 years, what was the age (in years) of the second student?
- (1) 22 years (2) 14 years
(3) 18 years (4) 36 years
46. Earnings of A and B are in the ratio 7 : 9 respectively and their expenditures are in the ratio 9 : 7 respectively. What is the ratio of their savings?
- (1) 1 : 7 (2) 7 : 5
(3) 1 : 1 (4) Can't be determined
47. Three friends A, B and C went on a picnic. A brought 5 apples and B brought 3 apples, and C had Rs. 80. They divided the apples equally among themselves and C gave all his money to A and B for their contributions. What are the respective shares of A and B from that money?
- (1) 50, 30 (2) 70, 10
(3) 40, 40 (4) 33.33, 66.66
48. In a school, 20% of the boys are same in number with $\frac{1}{3}$ rd of the girls. What is the ratio of number of boys to that of girls in that school?
- (1) 3 : 5 (2) 2 : 3
(3) 5 : 3 (4) 3 : 2
49. In two alloys, respective ratios of copper and zinc are 4 : 1 and 1 : 3. 10 kg of the 1st alloy, 16 kg of the 2nd alloy and some quantity of pure copper are melted together. In the resultant alloy the ratio of copper to zinc is 3 : 2. Find the weight (in kg) of the new alloy.
- (1) 45 (2) 40
(3) 35 (4) 50
50. A invested Rs.4,500 in a business. After few months, B joined him with Rs.3,000. At the end of the year, the total profit was divided between them in the ratio 3 : 1 respectively. After how many months did B join the business?
- (1) 2 months (2) 4 months
(3) 6 months (4) 8 months
51. The ratio of number of chocolates with Munni and Chameli is 3 : 2. If Chameli has 5 chocolates less than Sheela who has three more chocolates than Munni, find the total number of chocolates with three of them.
- (1) 19 (2) 20
(3) 23 (4) 24

52. A sum of Rs.3,115 is distributed among A, B and C such that, if Rs.25, Rs.28 and Rs.52 be diminished from their shares respectively, the remainders shall be in the ratio 8 : 15 : 20. Find the difference in the shares of A and C.
 (1) Rs.1,452 (2) Rs.867
 (3) Rs.587 (4) None of these
53. The ratio of the present age of a mother to that of her daughter is 9 : 4. Ten years ago, the ratio of their ages was 7 : 2. What are their present ages?
 (1) 54 years, 24 years (2) 45 years, 20 years
 (3) 54 years, 28 years (4) 36 years, 16 years
54. A, B and C enter into a partnership. A invests Rs.800 for 4 months, B invest Rs.1,200 for 8 months and C invest Rs.1,400 for 10 months. They gain Rs.1,340 together. Find the share of A.
 (1) Rs.160 (2) Rs.700
 (3) Rs.480 (4) Rs.320
55. A bag contains Rs. 600 in the form of one rupee, 50 paise and 25 paise coins in the ratio of 3 : 4 : 12. Find the total value (in Rs.) of the 25 paise coins present in the bag.
 (1) Rs. 150 (2) Rs. 225
 (3) Rs. 200 (4) Rs. 75
56. Pawan and Prabjot are partners in a business. Pawan invests Rs.35,000 for 8 months and Prabjot invests Rs.42,000 for 10 months. Out of the profit of Rs.31,570, Pawan's share is
 (1) Rs.3,157 (2) Rs.18,942
 (3) Rs.6,314 (4) Rs.12,628
57. If $\frac{a}{2} = \frac{b}{3} = \frac{c}{4} = \frac{2a-3b+5c}{k}$, where a, b and c are not equal to zero, then 'k' is
 (1) 15 (2) 17
 (3) 13 (4) Can't be determined
58. Product M is produced by mixing chemical X and chemical Y in the ratio of 5 : 4. Chemical X is prepared by mixing two raw materials, A and B, in the ratio of 1 : 3. Chemical Y is prepared by mixing raw materials, B and C, in the ratio of 2 : 1. Then the final mixture is prepared by mixing 864 units of product M with water. If the concentration of the raw material B in the final mixture is 50%, how much water was added to product M?
 (1) 328 units (2) 368 units
 (3) 392 units (4) 616 units
59. Two identical vessels are filled with alcohol. From the first vessel 'a' litre of the solution is taken out and replaced with 'a' litre of water. From the resulting mixture 'a' litre of the mixture is removed and again replaced with 'a' litre of water. Same operation is done in the second vessel the same number of times, but the amount of solution removed and replaced with water is '2a' litres. What fraction of the volume of the vessels is 'a' if the strength of alcohols in the two vessels finally is in the ratio 25 : 16?
 (1) 1 : 5 (2) 12 : 38
 (3) 488 : 512 (4) 512 : 488
60. A bag contains 25 paise, 10 paise and 5 paise coins in the ratio of 1 : 2 : 3 respectively. If their total value is Rs.45, then number of 10 paise coins is
 (1) 75 (2) 150
 (3) 225 (4) 175
61. If $a - b : b - c : c - d = 1 : 2 : 3$, then what is the ratio of $(a + d) : c$?
 (1) 1 : 2 (2) 2 : 1
 (3) 3 : 1 (4) None of these
62. Some amount of acid is taken out of a 54 L vessel full of acid and then an equal amount of water is added to it. The process is repeated once more. As a result, the vessel now contains 24 L pure acid. How much of the acid was taken out initially (in L)?
 (1) 18 L (2) 27 L
 (3) 15 L (4) 36 L
63. What sum of money is divided between A, B and C, if B and C together get Rs.100, C and A together get Rs.50, and A gets twice as much as B gets?
 (1) Rs.200 (2) Rs.225
 (3) Rs.240 (4) Rs.250
64. One-litre can is full with a mixture of water and alcohol, and alcohol content is 20% of the total mixture. If 10% of this liquid is taken out and pure alcohol is added to fill the can, what will be the percentage of alcohol in the new mixture?
 (1) 28% (2) 30%
 (3) 27% (4) 18%

Class Practice Exercise 12

Percentages I

DIRECTIONS: For the following questions write your answer in the space provided.

1. 11% of 15500 =
 (1) 1705 (2) 1710
 (3) 1695 (4) 1725
2. Two tailors P and Q are paid a total of \$ 550 per week by their employer. If P is paid 120 percent of the sum paid to Q, how much is Q paid per week?
 (1) \$ 200 (2) \$ 300
 (3) \$ 250 (4) None of these
3. 13% of 1700 =
 (1) 221 (2) 221.5
 (3) 219 (4) 228
4. In a maternity centre, 5% of all the childbirth cases results in twins. What is the approximate percentage of twins out of the total children born?
 (1) 9.52% (2) 4.76%
 (3) 5% (4) 10%
5. Three friends contributed to a charity fundraiser and collected ₹1136, ₹7636, and ₹11,628 respectively. What percentage of the total amount collected was contributed by the friend who donated the most?
 (1) 57% (2) 60%
 (3) 65% (4) 90%
6. 17% of 5000 =
 (1) 850 (2) 800
 (3) 1700 (4) 8500
7. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:
 (1) 3200 (2) 3000
 (3) 2800 (4) 2700
8. 19% of 9990 =
 (1) 1898.1 (2) 1899
 (3) 189.81 (4) 1908
9. A student multiplied a number by 3 instead of 5. What is the percentage error in the calculation?
 (1) 34% (2) 40%
 (3) 54% (4) 64%
10. 23% of 7000 =
 (1) 1610 (2) 1690
 (3) 1530 (4) 16100
11. Two numbers A and B are such that the sum of 5% of P and 4% of Q is two-third of the sum of 6% of P and 8% of Q. Find the ratio of P : Q
 (1) 2 : 3 (2) 1 : 1
 (3) 3 : 4 (4) 4 : 3
12. 29% of 15000 =
 (1) 4320 (2) 4350
 (3) 4410 (4) 4500
13. Aarav jogs on a circular track for 1.5 hours, taking 't' hours per lap. He then cycles for '2t' hours, completing 17 laps in total. On another day, he jogs 2.5 hours and rides a scooter for '4t' hours, completing 50 laps. If cycling and scooter speeds are in the ratio 1:3, find the time to complete one lap on the scooter.
 (1) 40 sec (2) 50 sec
 (3) 60 sec (4) 60 sec
14. In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is equal to the number of students of 8 years of age which is 48. What is the total number of students in the school?
 (1) 72 (2) 80
 (3) 120 (4) 150
15. 31% of 36000 =
 (1) 11000 (2) 11160
 (3) 11200 (4) 11300
16. If 20% of P = Q, then Q% of 20 is the same as -
 (1) 5% of P (2) 20% of P
 (3) 4% of P (4) None of these
17. 37% of 240 =
 (1) 85 (2) 88.8
 (3) 90 (4) 92
18. If P = a% of b and Q = b% of a, then which of the following is true?
 (1) P is smaller than Q
 (2) P is greater than Q
 (3) Relationship between P and Q cannot be determined.
 (4) If a is smaller than b, then P is greater than Q
19. 41% of 8900 =
 (1) 3600 (2) 3640
 (3) 3649 (4) 3700
20. From the integers 1 through 70, what fraction (expressed as a percentage) end with a units' digit of either 1 or 9?
 (1) 1 (2) 14
 (3) 20 (4) 21
21. 43% of 5700 =
 (1) 2400 (2) 2450
 (3) 2451 (4) 2500
22. A warehouse stored a batch of electronic gadgets. After dispatching 40% of them to retailers, 420 gadgets remained in stock. How many gadgets were there initially?
 (1) 588 apples (2) 600 apples
 (3) 672 apples (4) 700 apples
23. 47% of 63000 =
 (1) 29500 (2) 29600
 (3) 29610 (4) 29700

24. Two athletes competed in a fitness test. One scored 9 points more than the other, and his score was 56% of the combined points of both athletes. What were their individual scores?
 (1) 42, 33 (2) 45, 30
 (3) 50, 25 (4) 38, 37
25. 53% of 18000 =
 (1) 9500 (2) 9530
 (3) 9540 (4) 9600
26. A researcher stores two identical chemical samples worth ₹5,000 each under different growth processes for 2 years. One sample grows at 10% per year, while the other grows at 10% every half-year. What is the difference in their values after 2 years?
 (1) Rs. 26.50 (2) Rs. 27.50
 (3) Rs. 25.50 (4) Rs. 24.50
27. A contestant receives 84% of all votes cast and wins by a margin of 476. How many votes were cast in total? (Assume that only two contestants are there)
 (1) 567 (2) 2975
 (3) 800 (4) 700
28. A gamer scored 110 points in a challenge, of which 3 were bonus hits worth 4 points each and 8 were power-shots worth 6 points each. What percentage of his total score came from regular moves?
 (1) 45% (2) 45.45%
 (3) 55.55% (4) 55%
29. 59% of 58500 =
 (1) 34400 (2) 34500
 (3) 34515 (4) 34600
30. A company sells a product at a marked price that is 40% above its cost price. During a festival, it offers two successive discounts of x% and 20%, but still makes a profit of 12%. Later, for another scheme, the company reduces the marked price by 10% and then offers a single discount of 25%, which results in a loss of 4%. What is the value of x?
 (1) 0% (2) 4%
 (3) 8% (4) 12%
31. 61% of 61400 =
 (1) 37400 (2) 37450
 (3) 37454 (4) 37500
32. A student has to obtain 33% of the total marks to pass. He got 125 marks and failed by 40 marks. The maximum marks are
 (1) 300 (2) 500
 (3) 800 (4) 1000
33. 67% of 144000 =
 (1) 96000 (2) 96400
 (3) 96480 (4) 97000
34. In a survey, 2% chose A and 60% chose B. The rest were unsure. If those unsure were 720 fewer than those choosing B, find the total number of people surveyed.
 (1) 3600 (2) 1440
 (3) 1800 (4) Data Inadequate
35. 71% of 56000 =
 (1) 39500 (2) 39600
 (3) 39760 (4) 40000
36. In an examination, 35% of the students passed and 455 failed. How many students appeared for the examination?
 (1) 490 (2) 700
 (3) 845 (4) 1300
37. 73% of 8,70,000 =
 (1) 634,000 (2) 635,000
 (3) 635,100 (4) 636,000
38. A person who spends $66\frac{2}{3}\%$ of his income is able to save Rs. 1200 per month. His monthly expenses (in Rs.) are –
 (1) Rs. 1200 (2) Rs. 2400
 (3) Rs. 3000 (4) Rs. 3200
39. 79% of 14000 =
 (1) 11000 (2) 11050
 (3) 11060 (4) 11100
40. The rate of inflation in country is 100 % p.a. Cost of an Article is 6\$ now then what will be the cost of the article two year from now? (Inflation means general increase in Price level)
 (1) \$18 (2) \$20
 (3) \$24 (4) \$30
41. What is the percentage increase in the area of a rectangle if its length is increased by 20%?
 (1) 44% (2) 40%
 (3) 20% (4) $16\frac{2}{3}\%$
42. 83% of 24000 =
 (1) 19800 (2) 19900
 (3) 19920 (4) 20000
43. After deducting a commission of 5% on TV set costs Rs. 9595. Its marked price is –
 (1) Rs. 10,000 (2) Rs. 10,075
 (3) Rs. 10,100 (4) Rs. 10,500
44. 89% of 11000 =
 (1) 9750 (2) 9780
 (3) 9790 (4) 9800
45. Rohit decides to invest a certain amount of money in savings account of a bank that pays 20% compound interest per annum for a period of two years. What is the sum of money that Rohit should invest so that he receives Rs. 200 at the end of first year, and Rs. 400 at the end of second year?
 (1) Rs. 500 (2) Rs. 466.66
 (3) Rs. 155.55 (4) Rs. 2000
46. 97% of 13000 =
 (1) 12600 (2) 12605
 (3) 12610 (4) 12650
47. Gauri went to the stationers and bought things worth Rs. 25, out of which 30 paise went on sales tax on taxable purchase. If the tax rate was 6%, then what was the cost of tax-free items?
 (1) Rs. 15 (2) Rs. 15.70
 (3) Rs.19.70 (4) Rs. 20
48. 111% of 5600 =
 (1) 6200 (2) 6210
 (3) 6216 (4) 6220

49. The price of a car is Rs. 3, 25,000. It was insured to 85% of its price. The car was damaged completely in an accident and the insurance company paid 90% of the insurance. What was the difference between the price of the car and the amount received?
- (1) Rs. 32,500 (2) Rs. 48,750
(3) Rs. 76,375 (4) Rs. 81,250
50. 137% of 4000 =
- (1) 5450 (2) 5475
(3) 5480 (4) 5500
51. In a competitive examination in State A, 6% candidates got selected from the total appeared candidates. State B had an equal number of candidates appeared and 7% candidates got selected with 80 more candidates got selected than B. What was the number of candidates appeared for each State?
- (1) 7600 (2) 8000
(3) 8400 (4) Data Inadequate
52. 215% of 8000 =
- (1) 17,000 (2) 17,100
(3) 17,200 (4) 17,500
53. Find the approximate percent value of fraction: $\frac{338}{473}$
- (1) 70.4 (2) 69.4
(3) 71.2 (4) 74.3
54. When 15% is lost in grinding wheat, a country can export 30 lakh tons of wheat. On the other hand, if 10% is lost in grinding, it can export 40 lakh tons of wheat. The production of wheat in the country is
- (1) 20 lakh tons (2) 80 lakh tons
(3) 200 lakh tons (4) 800 lakh tons
55. 167% of 7500 =
- (1) 12,525 (2) 12,52,5
(3) 12,675 (4) 12,750
56. A tempo is insured to the extent of $\frac{4}{5}$ of its original value. If the premium on it at the rate of 1.3 percent amounts to Rs. 910, the original value of tempo is
- (1) Rs. 78,500 (2) Rs. 80,000
(3) Rs. 82,500 (4) Rs. 87,500
57. 56% is equivalent to
- (1) 0.56 (2) $\frac{14}{25}$
(3) $\frac{56}{100}$ (4) All of the above
58. A student multiplied a number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation?
- (1) 34% (2) 44%
(3) 54% (4) 64%
59. 4% is equivalent to
- (1) 0.04 (2) $\frac{4}{100}$
(3) $\frac{1}{25}$ (4) All of the above
60. The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is
- (1) 4.37% (2) 5%
(3) 6% (4) 8.75%
61. 0.6% is equivalent to
- (1) 0.006 (2) $\frac{6}{1000}$
(3) $\frac{3}{500}$ (4) All of the above
62. Three candidates contested an election and received 1136, 7636 and 11628 votes respectively. What percentage of total votes did the winning candidate get?
- (1) 56% (2) 57%
(3) 58% (4) 59%
63. 0.08% is equivalent to
- (1) 0.008 (2) $\frac{8}{1000}$
(3) $\frac{1}{125}$ (4) All of the above
64. Two numbers A and B are such that the sum of 5% of A and 4% of B is two-third of the sum of 6% of A and 8% of B. Find the ratio of A : B.
- (1) 3 : 4 (2) 4 : 3
(3) 2 : 3 (4) 3 : 2
65. 6% is equivalent to
- (1) 0.06 (2) 0.6
(3) 0.006 (4) 6
66. If one number is 80% of the other and 4 times the sum of their squares is 656, then the numbers are –
- (1) 4, 5 (2) 8, 10
(3) 16, 20 (4) None of these
67. 28% is equivalent to
- (1) 0.28 (2) 2.8
(3) 28 (4) 0.028
68. When any number is divided by 12, then dividend becomes $\frac{1}{4}$ th of the other number. By how much percent first number is greater than the second number?
- (1) 150 (2) 200
(3) 300 (4) Data inadequate
69. 0.2% is equivalent to
- (1) $\frac{2}{100}$ (2) $\frac{2}{1000}$
(3) $\frac{1}{50}$ (4) None of these
70. The difference of two numbers is 20% of the larger number. If the smaller number is 20, then the larger number is
- (1) 25 (2) 45
(3) 50 (4) 80
71. From 1998 to 1999 the percentage increase in male population was 20% and from 1999 to 2000 it decreases by 10% and Male population of a city in year 2000 is 1.08 million. Then what was the male population of the city in 1998?
- (1) 0.95 mn (2) 1 mn
(3) 1.05 mn (4) 1.1 mn
72. The percentage rate at which a sum becomes
Quantity A: 4 times of itself in 15 years at simple interest
Quantity B: 20%
- (1) if quantity in column A is greater;
(2) if quantity in column B is greater;
(3) if quantities in both the columns are equal
(4) if the relationship cannot be determined from the information given.

73. Ravi and Sameer can complete a job in 40 and 50 days respectively. They work only on weekdays (9 am–5 pm). They started on Monday, 20th Feb 2012. Sameer was absent from 14th–18th March. When was the job completed?

- (1) 22nd March 2012 (2) 23rd March 2012
(3) 24th March 2012 (4) 25th March 2012

74. At what percentage above the cost price must the price of an article be marked, so as to gain more than 20% after offering a discount of 25%. (Indicate all the correct choices)

- (1) 20% (2) 50%
(3) 60% (4) 70%
(5) 80%

75. Rajeev buys good worth Rs. 6650. He gets a rebate of 6% on it. After getting the rebate, he pays sales tax @ 10%. Find the amount he will have to pay for the goods.

- (1) Rs. 6876.10 (2) Rs. 6999.20
(3) Rs. 6654 (4) Rs. 7000

76. What percent is 1.25 of 10?

- (1) 12% (2) 12.5%
(3) 13% (4) 10%

77. Find the approximate value of the following fractions.

- $\frac{1.6 \times 1.12}{6.63}$
(1) 27.63% (2) 27.02%
(3) 28.11% (4) 26.53%

78. Alpha earns \$8000 and Beta earns 25% more than Alpha. Find salary of Beta?

- (1) \$9500 (2) \$10,000
(3) \$10,500 (4) \$11,000

79. In a town of population 1,20,000, 55% are males and rest are females. If 48% of males and 60% of females can vote, then what is the total number of voters in the town?

- (1) 64,320 (2) 64,080
(3) 70,000 (4) None of these

80. Salary of Lisa, in the year 2020 is \$10000. Next year, it will increase by 200%. What will be her salary in 2021.

- (1) \$20,000 (2) \$25,000
(3) \$30,000 (4) \$40,000

81. A bakery opens only when both Alice and Bob work. Alice works 4 days, rests 4 days, repeating. Bob works 5 days, rests 5 days, repeating. If they ran the bakery for 700 days, how many days was it open?

- (1) 172 (2) 176
(3) 184 (4) 196

82. Price of an iPhone in a sale was \$200, which was 20% Of your pocket money. What is your pocket Money?

- (1) \$800 (2) \$1000
(3) \$1200 (4) \$1500

83. Vikas marks up an item successively by x% and y% on his cost price. Then he gives a single discount of z% to bring the selling price down to the level of cost price. Which of the following is definitely true?

- (1) $x + y = z$ (2) $x + y > z$
(3) $x + y < z$ (4) Data inconsistent

84. In a Class, there were 200 students. In 3rd lecture, 48 students moved to conference hall. What % of students are still in the class?

- (1) 76% (2) 78%
(3) 80% (4) 82%

85. Find the approximate value of the following fractions.

- $\frac{547}{1973}$
(1) 29.82 (2) 27.72
(3) 28.92 (4) 26.92

86. Anant got 30% of the maximum marks and failed by 10 marks. If maximum marks are 150 then find passing marks?

- (1) 50 (2) 55
(3) 60 (4) 65

87. The price of petrol was increased by 25% and a person reduced his consumption by 25%. What is the percentage change in the expenditure incurred by him on petrol?

- (1) 6.25% decrease (2) 7% decrease
(3) 6.75% increase (4) 6% increase

88. In a class 40% are boys and remaining are girls. Among boys 20% passed the exam and among girls 40% passed the exam. What is the pass percentage of entire class?

- (1) 28% (2) 32%
(3) 36% (4) 40%

89. A company has 3 divisions: A, B, and C. Last year, their profits were in the ratio 5:7:8. This year, the profits of division A increased by 20%, division B decreased by 10%, and division C increased by 5%. After these changes, the total profit of the company this year is ₹1,08,000. What was the profit of division B last year?

- (1) ₹34,500 (2) ₹36,500
(3) ₹38,000 (4) ₹35,700

90. Price of an Article is \$50 in January. In the month of February, it becomes 1.2 times of its If. Find percentage Increase?

- (1) 15% (2) 20%
(3) 25% (4) 30%

91. A watermelon weighs 8.5 kgs. It contains 70% water by weight. However, once left to dry in the hot sun for 3 days, it loses 75% of its water content (by weight). What is the water content (by weight) percentage in the watermelon after it has been dried in sun for 3 days?

- (1) 5% (2) 17.4%
(3) 36.9% (4) 58.1%

92. Stock value of Prakash is 0.92 times of Hiren then stock value of Prakash is what percent less than that of Hiren?

- (1) 6% (2) 8%
(3) 10% (4) 12%

93. Bhawna lent part of Rs.10,000 to Myra, one of her friends at 8% SI for 4 years. She invested the remaining amount at 12% SI. Total income from interest after 4 years was Rs. 3,800. What was sum (in Rs.) lent to Myra?

- (1) Rs. 5000 (2) Rs. 6250
(3) Rs. 7500 (4) Rs. 8000

94. Salary of A becomes $\frac{9}{8}$ times of his previous salary. By how much percent his salary has been increased?
- (1) 10% (2) 12%
(3) 12.5% (4) 15%
95. In a two-party election, votes against Bahumukhi Party rose by 25% from the previous election, and it lost by a margin twice its earlier winning margin. If 2,60,000 people voted each time, how many voted for Bahumukhi Party previously?
- (1) 1,10,000 (2) 1,50,000
(3) 1,40,000 (4) 1,20,000
96. When someone increases salary of Hitesh by 83.33 % then it becomes 121k then find the original salary of Hitesh?
- (1) 60,000 (2) 66,000
(3) 70,000 (4) 75,000
97. Narayan and Murthy invested some money at 6% and 7% per annum respectively of SI. At the end of 2 years, they found that together they received Rs. 354 as interest. One-fourth of Narayan's initial investment is equal to one-fifth of the money invested by Murthy. Find the total money invested.
- (1) Rs. 2,746.5 (2) Rs. 2,600
(3) Rs. 2,700 (4) Rs. 2,880
98. Donut company increases his production each month by 100 counts. In the Month of January 2020, the production was 800k. In which month of the year 2020, percentage increase in production will be lowest?
- (1) January (2) June
(3) December (4) September
99. In a colony, 60% are men and 12% are literate. If 20% of women are literate, find the percentage of illiterate men among all men. [Assume, that every person in the colony is either literate or illiterate]
- (1) 80% (2) 83.33%
(3) 92% (4) 93.33%
100. 12% is what % less than 16%?
- (1) 20% (2) 25%
(3) 30% (4) 33.33%
101. Find the approximate percent value of the fractions:
- $\frac{8}{19}$
- (1) 41.02 (2) 40.08
(3) 41.08 (4) 42.08
102. Akshay and Anand have their salary in the ratio of 5:8 then Anand's salary is what percent more than that of Akshay?
- (1) 50% (2) 55%
(3) 60% (4) 65%
103. In accordance with the provisions of the new budget, Vaibhav pays tax at the rate of 5% of the taxable income and 10% as special surcharge on the amount of income tax levied. He altogether pays a tax of Rs.22,000. Find his gross pay. (Taxable income = 80% of Gross Pay)
- (1) Rs.6,00,000 (2) Rs.5,00,000
(3) Rs.4,00,000 (4) Rs.3,50,000
104. Amit has 12.5% share value less than Bhavesh, then Bhavesh's share value is what percent more than that of Amit?
- (1) 12.5% (2) 14.29%
(3) 15% (4) 16%
105. If the price of petrol increases by 25%, then by how much percent a user cut down his consumption so that his expenditure on petrol remains constant?
- (1) 15% (2) 20%
(3) 25% (4) 30%

Class Practice Exercise 13

Percentages II

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. Three numbers are such that the first number is 30% of third number and second number is 40% of the third number. First number is what percent of the second number?
 (1) 133.33% (2) 25%
 (3) 33.33% (4) 75%
2. Express each of the following as a decimal: 0.04%
 (1) 0.04 (2) 0.004
 (3) 0.0004 (4) 0.00004
3. If 25% of a number is subtracted from a second number, the second number reduces to its five-sixth. What is the ratio of the first number to the second number?
 (1) 1:3 (2) 2:3
 (3) 3:2 (4) Data inadequate
4. Express each of the following as rate percent. $\frac{23}{36}$
 (1) 63.11% (2) 63.88%
 (3) 65.66% (4) 61.44%
5. The price per unit of an article decreases by 4%, and the consumption (in units) decreases by 8%. The expenditure would decrease by
 (1) 12% (2) 12.32%
 (3) 11.68% (4) 4.32%
6. Express each of the following as rate percent. $6\frac{3}{4}$
 (1) 6.75% (2) 67.5%
 (3) 675% (4) 0.0675%
7. The sum of two numbers is $\frac{28}{25}$ of the first number. The second number is what percent of first?
 (1) 12% (2) 14%
 (3) 16% (4) 18%
8. Express each of the following as rate percent. 0.004
 (1) 4% (2) 0.4%
 (3) 0.04% (4) 0.00004%
9. Amala and Bindu started a pet service. After 6 months, Amala left and pet strength dropped by 50%. Bindu stayed for the full year. If the total profit is ₹30,000, what is Amala's share if profit is divided by (Pets $\times$ Time)?
 (1) Rs. 10,000 (2) Rs. 15,000
 (3) Rs. 20,000 (4) Rs. 5,000
10. 28% of 450 + 45% of 280 =
 (1) 126 (2) 252
 (3) 280 (4) 504
11. The sum of two numbers is 2490. If 6.5% of one number is equal to 8.5% of the other, then the number are –
 (1) 989, 1501 (2) 1011, 1479
 (3) 1401, 1089 (4) 1411, 1079
12. $16\frac{2}{3}\%$ of 600 gm - $33\frac{1}{3}\%$ of 180 gm =
 (1) 160 gm (2) 40 gm
 (3) 20 gm (4) 80 gm
13. Vaibhav has 4 equal passages with 5, 8, 8, and 6 questions (27 total). Reading one passage takes the same time as answering 12 questions. If his answering speed is constant, how much must he increase his reading speed to reduce his total exam time by 10%?
 (1) 20% (2) 18.5%
 (3) 25% (4) 9.8%
14. 2 is what percent of 50?
 (1) 4% (2) 8%
 (3) 12% (4) 16%
15. In a cricket match India beat Australia by 2 runs. If Sachin scores % of the total score of India then which of the following can be the score of India? (Mark all the correct answers)
 (1) 306 (2) 308
 (3) 317 (4) 328
 (5) 351 (6) 389
16. $\frac{1}{2}$ is what percent of $\frac{1}{3}$?
 (1) 50% (2) 66.66%
 (3) 150% (4) 166.66%
17. Which of the following multipliers will cause a number to be increased by 29.7%?
 (1) 1.297 (2) 12.97
 (3) 129.7 (4) 1297
18. What percent of 7 is 84?
 (1) 12% (2) 120%
 (3) 1200% (4) 8.33%
19. A solution of sugar and water contains 20% sugar by weight. From this solution, if 25 kg of water is evaporated, the mixture now contains 25% of sugar. Find the quantity of sugar in the solution initially.
 (1) 50 kg (2) 125 kg
 (3) 100 kg (4) 25 kg
20. What percent of 2 metric tons is 40 quintals?
 (1) 20% (2) 50%
 (3) 200% (4) 2000%
21. A number when 35 is subtracted from it, reduces to its 80 percent. What is four-fifth of that number?
 (1) 70 (2) 90
 (3) 120 (4) 140
22. What percent of 6.5 litres is 130 ml?
 (1) 0.2% (2) 2%
 (3) 20% (4) 5%
23. If the price of a bicycle is increased by 9.09%, then by 8.33% and then by 7.7%, the price becomes Rs. 1,274. What is the original price of the bicycle?
 (1) Rs. 999 (2) Rs. 1,001
 (3) Rs. 1,100 (4) Rs. 1,150
24. Find the missing term: ?% of 25 = 2.125
 (1) 0.85 (2) 8.5
 (3) 53.125 (4) 85

25. If 75% of a number is added to 75, then the result is the number itself. The number is
 (1) 50 (2) 60
 (3) 300 (4) 400
26. Find the missing term: 9% of ? = 63
 (1) 7 (2) 70
 (3) 700 (4) 7000
27. Nala and Bheem ran a year-long catering service. Bheem managed it for 4 months more than Nala. Nala invested Rs. 20,000 more than Bheem. After paying the manager a Rs. 1,000 monthly salaries, they split the remaining Rs. 24,000 profit equally. Find Bheem's capital.
 (1) Rs. 60,000 (2) 140,000
 (3) Rs. 20.00 (4) Can't determine
28. Find the missing term: 0.25% of ? = 0.04
 (1) 0.16 (2) 1.6
 (3) 16 (4) 160
29. If a number x is 10% less than another number y and y is 10% more than 125, then x is equal to
 (1) 123.75 (2) 140.55
 (3) 143 (4) 150
30. Change following fractions in percentages $\frac{283}{1983} =$
 (1) 12.7% (2) 14.3%
 (3) 13.6% (4) 15.1%
31. A vendor sells 30% of his fruits and throws away 40% of the rest of the fruits. Next day he sells 50% of the remainder fruits and throws away the rest. What percent of the fruits does the vendor throw?
 (1) 51% (2) 49%
 (3) 63% (4) 72%
32. Change following fractions in percentages $\frac{1132}{1069} =$
 (1) 1.06 (2) 1.08
 (3) 1.04 (4) 1.02
33. By how much percent is four-fifth of 70 lesser than five-seventh of 112?
 (1) 24% (2) 30%
 (3) 36% (4) 42%
34. Change following fractions in percentages $\frac{783}{1869} =$
 (1) 43.2% (2) 41.9%
 (3) 27.2% (4) 40.6%
35. If a dealer gets a commission of 10% on the list price from the company, then the profit made by the company is 50% of its manufacturing cost, if the dealer's commission is increased to 25%, then what will be the profit percentage on the cost of manufacturing for the company?
 (1) 10% (2) 25%
 (3) 30% (4) 15%
36. Change following fractions in percentages $\frac{1234.5}{15.5} =$
 (1) 80.23% (2) 68.32%
 (3) 76.65% (4) 79.6%
37. What percentage of numbers from 1 to 70 have squares that end in the digit 1?
 (1) 1 (2) 14
 (3) 20 (4) 21
38. Change following fractions in percentages $\frac{4568}{238} =$
 (1) 18.4 (2) 17.8
 (3) 19.2 (4) 20.1
39. If sides of a square are increased by 30%, then what is the percentage increase in its area?
 (1) 9% (2) 30%
 (3) 60% (4) 69%
40. Change following fractions in percentages $\frac{987}{263} =$
 (1) 3.75 (2) 3.59
 (3) 3.93 (4) 3.32
41. The number which exceeds 16% of it by 42 is
 (1) 50 (2) 52
 (3) 58 (4) 60
42. Change following fractions in percentages $\frac{12.71}{29.5} =$
 (1) 41.26% (2) 45.35%
 (3) 47.38% (4) 43.08%
43. In 2009-10, items P, Q, and R averaged Rs. 64cr in sales. P and Q each sold half as much as R. In 2010-11, Q was replaced by S, raising the average to Rs. 80cr. P's sales grew by 30% and R's by 20%. Find S's sales in 2010-11.
 (1) the same as the sales of P in 2010-11.
 (2) the same as the sales of R in 2010-11.
 (3) Rs. 1.6 crore less than the average sales of the three items in 2009-10.
 (4) Both (1) and (3)
44. What is greatest in $16\frac{2}{3}\%$, $\frac{2}{15}$ and 0.17?
 (1) $16\frac{2}{3}\%$ (2) $\frac{2}{15}$
 (3) 0.17 (4) All are equal
45. If 35% of a number is 12 less than 50% of that number, then the number is
 (1) 40 (2) 50
 (3) 60 (4) 80
46. If the sales tax be reduced from $3\frac{1}{2}\%$ to $3\frac{1}{3}\%$, then what difference does it make to person who purchases an article with marked price of Rs. 8400?
 (1) Rs. 10 (2) Rs. 14
 (3) Rs. 18 (4) Rs. 20
47. In an election between two candidates, the candidate who gets 30% of the votes polled is defeated by 15,000 votes. The number of votes polled in favour of the winning candidate is
 (1) 11,250 (2) 15,000
 (3) 26,250 (4) 37,500
48. An inspector rejects 0.08% of the meters as defective. How many will be examined to reject 2?
 (1) 250 (2) 2500
 (3) 25000 (4) 5000

49. Subtracting 40% of a number from the number, we get the result as 30. The number is
 (1) 28 (2) 50
 (3) 52 (4) 70
50. Sixty-five percent of a number is 21 less than four-fifth of that number. What is the number?
 (1) 70 (2) 140
 (3) 150 (4) 210
51. In a class, 70% of the students are boys and 14% of the students play cricket. If 10% of boys play cricket, then what percentage of the cricket players are boys?
 (1) 80% (2) 70%
 (3) 60% (4) 50%
52. Difference of two numbers is 1660. If 7.5% of one number is 12.5% of the other number, find the two numbers.
 (1) 4150, 2490 (2) 2490, 830
 (3) 3320, 1660 (4) 4150, 2500
53. If 15% of 40 is greater than 25% of a number by 2, then the number is
 (1) 12 (2) 16
 (3) 24 (4) 32
54. In expressing a length 81.472 km as nearly as possible with three significant digits, find the percentage error.
 (1) 0.034% (2) 0.088%
 (3) 0.028% (4) 0.34%
55. Arun's salary has two parts - the basic and the dearness allowance. His salary is \$60,000, of which dearness allowance is $16\frac{2}{3}\%$. His basic is expected to go up by Rs. 2000 each year and his dearness allowance is expected to go up by Rs. 500 each year. After how many years will his expected salary be Rs. 77,500?
 (1) 6 (2) 7
 (3) 8 (4) 9
56. In an election between two candidates, 75% of the voters cast their votes, out of which 2% of the votes were declared invalid. A candidate got 9261 votes which were 75% of the total valid votes. Find the total number of votes enrolled in that election.
 (1) 15600 (2) 16000
 (3) 16800 (4) 18000
57. The difference between a number and its two-fifth is 510. What is 10% of that number?
 (1) 12.75 (2) 85
 (3) 204 (4) None of these
58. Shobha's 75-question test had 10 arithmetic, 30 algebra, and 35 geometry problems. She got 70% of arithmetic, 40% of algebra, and 60% of geometry correct. If 60% total is required to pass, how many more questions did she need?
 (1) 5 (2) 7
 (3) 10 (4) 15
59. Gun powder contains 75% nitre, 10% of sulphur and the rest is charcoal by weight. If quantity of gunpowder is 24 kg, then what is the weight of charcoal?
 (1) 3.6 kg (2) 4.5 kg
 (3) 5 kg (4) 5.5 kg
60. If 50% of $(x - y) = 30\%$ of $(x + y)$, then what percent of x is y ?
 (1) 25% (2) 40%
 (3) 75% (4) 400%
61. Two-fifth of one-third of three-seventh of a number is 15. What is 40% of that number?
 (1) 72 (2) 84
 (3) 136 (4) 105
62. Mr. Jones gave 40% of the money he had, to his wife. He also gave 20% of the remaining amount to each of his three sons. Half of the amount now left was spent on miscellaneous items and the remaining amount of Rs. 12,000 was deposited in the bank. How much money did Mr. Jones had initially?
 (1) Rs. 80,000 (2) Rs. 90,000
 (3) Rs. 1,00,000 (4) Rs. 1,20,000
63. Rs. 2,189 is divided into three parts such that their amounts after 1, 2 and 3 years respectively may be equal. If the rate of simple interest being 4% p.a. in all cases, the smallest part is
 (1) Rs. 389 (2) Rs. 756
 (3) Rs. 703 (4) Rs. 352
64. 10% of the inhabitants of a village having died of cholera, a panic set in, during which 25% of the remaining inhabitants left the village. The population is then reduced to 4050. Find the number of original inhabitants.
 (1) 5400 (2) 5800
 (3) 6000 (4) 6750
65. If 35% of a number is 175, then what percent of 175 is that number?
 (1) 35% (2) 65%
 (3) 280% (4) None of these
66. A salesman's commission is 5% on all sales upto Rs. 10,000 and 4% on all sales exceeding this. He remits Rs. 31,100 to his parent company after deducting his commission. Find the total sales.
 (1) Rs. 30,000 (2) Rs. 31,500
 (3) Rs. 32,500 (4) Rs. 35,000
67. If all the sides of a cuboid are increased by 20%, then by what percentage does its volume increase?
 (1) 20% (2) 44%
 (3) 60% (4) 72.8%
68. Raman's salary was decreased by 50% and subsequently increased by 50%. How much percent does he lose?
 (1) 20% (2) 25%
 (3) 30% (4) 50%
69. If 120 is 20% of a number, then 120% of that number will be –
 (1) 20 (2) 120
 (3) 360 (4) 720
70. Paulson spends 75% of his income. His income is increased by 20% and he increased his expenditure by 10%. Find the percentage increase in his savings.
 (1) 50% (2) 40%
 (3) 10% (4) 6%

71. A and B are two friends, each having at least a rupee. If A gives B a sum of Rs. 20, then A has 40% of the amount with B. If B gives A, a sum of Rs. 40, then B will have 40% of the amount with (1) What is the amount (in Rs.) with A initially?
- (1) Rs. 60 (2) Rs. 80
(3) Rs. 52 (4) Rs. 100
72. If the salary of Ram is 33.33% more than Laxman's, then by how much percent is Laxman's salary less than that of Ram?
- (1) 20 % (2) 25 %
(3) 33.33% (4) 12.5 %
73. In a test, Anil got 30% of the maximum marks and failed by 24 marks. In the same test, Chetan's got 38% of the maximum marks and got 8 marks more than the pass mark. Find pass marks
- (1) 144 (2) 120
(3) 152 (4) 400
74. A's share in profit is 25% more than that of B, whose share is 20% more than that of (3) By how much percent is A's share greater than C's?
- (1) 20.35% (2) 33.33 %
(3) 40.54 % (4) None of these
75. Two lumps having mixture of gold and silver together weighs 20 kg. One lump contains 75% gold and 31.25 gm silver per kg. Another lump contains 85% gold and 30 gm silver per kg. Total quantity of silver in two lumps is 617.5 gm. If two lumps are melted to form one, then percentage of gold in it will be
- (1) 75% (2) 87%
(3) 15.6% (4) 78%
76. 12% of a certain sum of money is Rs. 43.5. Find the amount.
- (1) 342.5 (2) 362.5
(3) 462.5 (4) 402.5
77. In an exam, a student scored 50% of the maximum marks and yet failed by 12 marks. If he had scored 10% more than what he scored, he would have just managed to get the pass percentage. What are the maximum marks of the paper?
- (1) 120 (2) 300
(3) 200 (4) 240
78. Two numbers are respectively 20% and 50% more than a third. What percent is the first of the second?
- (1) 60% (2) 75%
(3) 80% (4) 50%
79. 20% of x is equal to 10.
- (1) 50 (2) 2
(3) 200 (4) 5
80. Ram's salary increased from Rs. 630 to Rs. 700. Find the percentage increase.
- (1) 11.11% (2) 9.09%
(3) 10% (4) 8.33%
81. The quantity of tea produced by a company increased from 15000 tons to 19500 tons from 2006 to 2007. Find the percentage increase in the quantity of its tea production from 2006 to 2007.
- (1) 30% (2) 23.1%
(3) 45% (4) 30%
82. A man loses 12.5% of his money in tax. He spends 70% of the balance on household expenses and is left with Rs. 210. How much money did he have at first?
- (1) 1200 (2) 1500
(3) 1000 (4) 800
83. A college raised 75% of its building fund from 60% of potential donors, averaging Rs. 600 each. To reach 100% of the goal, what must be the average donation from the remaining 40% of donors?
- (1) Rs. 300 (2) Rs. 400
(3) Rs. 200 (4) Rs. 450
84. Due to fall in manpower, the production in a factory decreases by 25%. By what percent should the working hours be increased to restore the original production?
- (1) 20% (2) 25%
(3) 33.33% (4) 30%
85. In a basket of fruits, 60% are mangoes and remaining 40% are apples, 25% of the apples are green and the rest 75% are red. Of the mangoes, 80% are red and the rest of the mangoes are green. What percentage of the green fruits are mangoes?
- (1) 12% (2) 45.45%
(3) 22% (4) 54.54%
86. What percent is Rs. 1.25 of Rs. 10?
- (1) 25% (2) 12.25%
(3) 1.25% (4) 12.5%
87. 11 men working for 4 hours can make 1 widget of type P and 2 widgets of type Q. 14 men working for 7 hours can make 3 widgets of type P and 4 widgets of type Q. How many hours will 15 men take to make 2 widgets of type P and 5 widgets of type Q?
- (1) 7 (2) 6
(3) 12 (4) 8
88. A student must secure 40% marks to pass. If he gets 40 marks and fails by 40 marks, find the maximum marks.
- (1) 400 (2) 200
(3) 250 (4) 160
89. A's income is 70% of B's. B's income is 50% of C's. If C's income is Rs. 1,00,000, then what is A's income?
- (1) Rs. 35,000 (2) Rs. 50,000
(3) Rs. 70,000 (4) Rs. 60,000
90. The number of seats in the cinema hall is increased by 25% and price per ticket is decreased by 10% what will be the effect on overall sales?
- (1) 10% Decrease (2) 15% Increase
(3) 12.5% Increase (4) 10% decrease

91. In a certain town, people are either married or unmarried. Similarly, either they work or they don't work. 37% of the total population was married and 76% of the married people were working. Similarly, 54% of the unmarried people were working. If the total number of people who work was 1,24,280, then the total population of the town was:
- (1) 90,000 (2) 2,00,000
(3) 1,80,000 (4) 2,16,000
92. If, after giving the discount of 20%, Ram wants to earn the profit of 20%, then by how much percent should the price be marked higher?
- (1) 40 % more (2) 33.33 % more
(3) 50% more (4) 25 % more
93. If A's salary is 30% more than B's salary then B's salary is what percent less than A's salary?
- (1) 27.27% (2) 22.22%
(3) 48.46% (4) 23.07%
94. Two persons A & B bought "Land for Rs. 2 lacs" and "Flat for Rs. 2.2 lacs" respectively. The price of the flat rises by 10 % and that of Land rises by 20% every year. If they exchange their assets after two years find the % profit of the gainer.
- (1) 8.19 % (2) 4.34 %
(3) 12 % (4) 19 %
95. Find the approximate value of the following fractions. $\frac{5}{19}$
- (1) 24.3% (2) 25.7%
(3) 26.3% (4) 24.7%
96. When a certain sum is invested at 7% p.a. instead of 5% p.a. for 3 years under SI then there is increase of Rs720 in the SI.
- Quantity A:** Principal
Quantity B: 12000
- (1) if quantity in column A is greater;
(2) if quantity in column B is greater;
(3) if quantities in both the columns are equal
(4) if the relationship cannot be determined from the information given.
97. What percent of $7y/8$ is y ?
- (1) 112.5% (2) 114.28%
(3) 116.66% (4) 111.11%
98. A shopkeeper gains (Profit) on an average 20% by selling multiple articles in a month. During that month an article was sold at \$360.
- Quantity A:** Cost Price of that article
Quantity B: 300
- (1) if quantity in column A is greater;
(2) if quantity in column B is greater;
(3) if quantities in both the columns are equal
(4) if the relationship cannot be determined from the information given.
99. There is an alloy A of silver and copper. A certain weight of this alloy is mixed with 15 kg of pure silver and is melted. The new alloy B contains 90% of silver. If alloy A is mixed with 10kg of a 90% silver alloy, and is melted to form new alloy C, then C is found to contain 84% silver. Find the percentage of silver in A.
- (1) 80% (2) 90%
(3) 75% (4) 85%
100. By selling 22 articles, the selling price of 11 articles is lost.
- Quantity A:** Loss incurred on an article as a fraction of its cost price.
Quantity B: $1/2$
- (1) if quantity in column A is greater;
(2) if quantity in column B is greater;
(3) if quantities in both the columns are equal
(4) if the relationship cannot be determined from the information given.
101. An electrical contractor purchased a certain amount of wire, 10% of which was stolen. After using 90% of the remaining wire, he had 47.25 m of wire left. How much wire did he purchase?
- (1) 472.5 m (2) 425.25 m
(3) 525 m (4) 500 m
102. If radius of a circle increases by 40% then what is the % change in the Circumference and area of the circle?
- (1) 40% and 96% (2) 40% and 80%
(3) 20% and 40% (4) 40% and 160%
103. Find the approximate value of the following fractions.
- $\frac{443}{898}$
- (1) 48.5 (2) 49.8
(3) 49.4 (4) 47.9
104. In a car showroom 20% of the cars are red, 25% of the remaining are blue and the rest are black. If there are 720 black cars in the showroom then find the total number of cars in the showroom?
- (1) 1200 (2) 1309
(3) 1600 (4) 1000
105. A candidate who gets 20% marks fails by 10 marks but another candidate who gets 42% marks gets 12 marks more than the passing marks. Find the maximum marks.
- (1) 50 (2) 100
(3) 150 (4) None of these
106. When a positive number x is multiplied by the same number then resulting number is what percent more than the original number.
- (1) $(x-1)\%$ (2) $(x-1) \times 100\%$
(3) $(x+1)\%$ (4) $(x+1) \times 100\%$
107. Dinesh, Eswar, Girl, and Hari travel from A to B. Dinesh takes Eswar on his bike (60 km/hr) while Girl walks (10 km/hr) and Hari cycles. Dinesh drops Eswar, who finishes on foot (10 km/hr), and rides back to pick up Girl. All four reach B simultaneously. Find Hari's average speed.
- (1) $480/17$ (2) $600/23$
(3) $900/29$ (4) $540/19$
108. If 400 is increased by 20%, and then reduced by $y\%$, yielding 432 then find y ?
- (1) 10 (2) 8
(3) 12 (4) 11.1
109. If $X = 37.5\%$ of 20% of 48 and $Y = 14.28\%$ of 27.27% of 77, then
- (1) $X > Y$ (2) $X = Y$
(3) $X < Y$ (4) $X - Y = 1.4$

Class Practice Exercise 14

Percentages III

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. Machines M_1 , M_2 , and M_3 produce 40%, 30%, and 30% of a stockpile, respectively. Defect rates are 3% for M_1 and 1% for M_2 . For M_3 , 95% of products are non-defective. What is the total percentage of defective products in the stockpile?
 (1) 3% (2) 5%
 (3) 30% (4) 32.7%
2. The salary of a person was reduced by 10%. By what percent should his reduced salary be raised so as to bring it at par with his original salary?
 (1) 10% (2) 11%
 (3) 11.11% (4) 12.5%
3. $\sqrt{784} + ? = 78\%$ of 500
 (1) 342 (2) 352
 (3) 362 (4) 372
4. When the price of a product was decreased by 10%, the number sold increased by 30%. What was the effect on the total revenue?
 (1) 17% increase (2) 20% increase
 (3) 17% decrease (4) 13% increase
5. Vijay lent P_1 , P_2 , and P_3 for two years. X and Y paid 10% and 15% simple interest; Z paid 10% interest compounded annually. If total interest is Rs. 2415 and $20\% P_1 + 30\% P_2 = 1312.5$, find P_3 .
 (1) 5250 (2) 5600
 (3) 5450 (4) 5750
6. If the numerator of a fraction be increased by 15% and its denominator be diminished by 8%, the value of the fraction is $\frac{15}{16}$. Find the original fraction.
 (1) $\frac{5}{8}$ (2) $\frac{4}{5}$
 (3) $\frac{3}{4}$ (4) $\frac{7}{8}$
7. If A earns 16.66% more than B, how much percent does B earn less than A?
 (1) 15.5% (2) 16.66%
 (3) 12.5% (4) 14.28%
8. The value of a machine depreciates at the rate of 10% per annum. If its present value is Rs. 1,62,000, what will be its worth after 2 years? What was the value of the machine 2 years ago?
 (1) Rs. 1,31,220, Rs. 2,00,000
 (2) Rs. 1,29,600, Rs. 2,02,500
 (3) Rs. 1,31,220, Rs. 1,96,020
 (4) Rs. 2,00,000, Rs. 1,31,220
9. ABC and Company produces nuts for FGH Ltd. The rejection rate for the nuts produced was 4% for the first quarter of the month and 8% for the next quarter of the month. The monthly rejection rate was maintained at 7%. Thus, ratio of first quarter to second quarter production was:
 (1) 1 : 3 (2) 2 : 1
 (3) 1 : 1 (4) 1 : 2
10. During one year, the population of a town increased by 5% and during the next year, the population decreased by 5%. If the total population is 9975 at the end of the second year, then what was the population size in the beginning of the first year?
 (1) 9500 (2) 9975
 (3) 10,000 (4) 10,500
11. The population of a town is 1,76,400. It increases at the rate of 5% per annum, what will be its population 2 years hence and what was it 2 years ago?
 (1) 194,040, 160,364 (2) 160,000, 194,481
 (3) 194,481, 160,000 (4) 194881, 160000
12. If A earns $33\frac{1}{3}\%$ more than B, how much percent B earn less than A?
 (1) 20% (2) 33.33%
 (3) 25% (4) 28.57%
13. If, $\frac{1}{a+b} = \frac{1}{a} + \frac{1}{b}$, then the value of $a^3 - b^3$ will be
 (1) 0 (2) 5
 (3) $\frac{3}{2}$ (4) 1
14. If A's salary is 20% less than B's salary, by how much percent is B's salary more than A's?
 (1) 20% (2) 24%
 (3) 25% (4) 30%
15. In the new budget, the price of Kerosene oil rose by 20%. By how much percent must a person reduce his consumption so that his expenditure on it does not increase?
 (1) 20% (2) 15%
 (3) 16% (4) 16.66%
16. How many kg of pure salt must be added to 30 kg of 2% solution of salt and water to increase it to 10% solution?
 (1) 2.4 kg (2) $\frac{3}{8}$ kg
 (3) 3 kg (4) $\frac{8}{3}$ kg
17. A flat costs Rs. 2,75,000 cash, or Rs. 1,10,000 down plus two equal annual installments at 20% compound interest. If bought on installments, how much extra is paid compared to the cash price?
 (1) Rs. 55,000 (2) Rs. 51,000
 (3) Rs. 59,000 (4) Rs. 61,000

18. Due to reduction of $6\frac{1}{4}\%$ in the price of sugar, a man is able to buy 1 kg more for Rs. 120. Find the original and reduced rate of sugar.
 (1) Rs. 8.00/kg, Rs. 7.50/kg
 (2) Rs. 7.50/kg, Rs. 8.00/kg
 (3) Rs. 9.00/kg, Rs. 8.44/kg
 (4) Rs. 8.25/kg, Rs. 7.75/kg
19. In an election a candidate who gets 75% of the votes elected the winner and wins by a surplus of 644 votes. What is the total number of votes polled?
 (1) 859 (2) 1024
 (3) 2576 (4) 1288
20. In an examination, 35% of total students failed in Hindi, 45% failed in English and 20% in both. Find the percentage of those who passed in both the subjects.
 (1) 20% (2) 30%
 (3) 40% (4) 60%
21. If $\sqrt{4096} = 64$, then the value of $\sqrt{40.96} + \sqrt{0.4096} + \sqrt{0.004096} + \sqrt{0.0004096}$ will be
 (1) 7.09 (2) 7.1014
 (3) 7.1104 (4) 7.12
22. In an examination, 80% of the students passed in English, 85% in Mathematics and 75% in both English and Mathematics. If 40 students failed in both the subjects, find the total number of students.
 (1) 160 (2) 200
 (3) 400 (4) 320
23. A student has to obtain 33% of the total marks to pass. He got 150 marks and failed by 48 marks. The maximum marks are –
 (1) 450 (2) 500
 (3) 660 (4) 600
24. The ratio 5 : 4 expressed as a percent equals:
 (1) 12.5% (2) 40%
 (3) 80% (4) 125%
25. In an examination, 25% of the students passed and 555 failed. How many students appeared for the examination?
 (1) 700 (2) 793
 (3) 2220 (4) 740
26. A sum of Rs. 1800 was split into two parts. One of the parts was invested at 6% p.a. simple interest and the other part was invested at 5% p.a. compound interest. Both parts were invested for 2 years. If the total interest realized from the two parts is Rs. 195, find the part invested at simple interest.
 (1) Rs. 800 (2) Rs. 600
 (3) Rs. 1200 (4) Rs. 100
27. 3.5 can be expressed in terms of percentage as
 (1) 0.35% (2) 3.5%
 (3) 35% (4) 350%
28. If 74% of a number is 555, then what will be 44% of that number?
 (1) 750 (2) 330
 (3) 290 (4) 310
29. Half of 1 percent written as decimal is
 (1) 0.005 (2) 0.05
 (3) 0.02 (4) 0.2
30. If A's salary is 60% less than B's salary by how much percent is B's salary more than A's?
 (1) 60% (2) 90%
 (3) 100% (4) 150%
31. What is 15 percent of Rs. 34?
 (1) Rs. 3.40 (2) Rs. 3.75
 (3) Rs. 4.50 (4) Rs. 5.10
32. A sum was split into two equal parts. One of the parts was lent at R% p.a. simple interest and the other part was lent at R% p.a. compound interest. Both parts were lent for 3 years. The difference of the interest accrued on the two parts at the end of two years and at the end of three years are in the ratio 10 : 31. Find R.
 (1) 5 (2) 8
 (3) 10 (4) 2
33. 63% of $3\frac{4}{7}$ is
 (1) 2.25 (2) 2.40
 (3) 2.50 (4) 2.75
34. The price of an article increased by 30% and 20%. It is then decreased by 10%. If its final value is Rs. 280.80. Find its initial value.
 (1) Rs. 180 (2) Rs. 220
 (3) Rs. 200.50 (4) 200
35. 88% of 370 + 24% of 210 - ? = 118
 (1) 256 (2) 258
 (3) 268 (4) 358
36. The price of a book increased successively by 20% and 30%. If it finally cost Rs. 468, find its initial value.
 (1) Rs. 312 (2) Rs. 320
 (3) Rs. 360 (4) Rs. 300
37. A student obtained 95 marks out of 250 in mathematics. If passing percentage is 42%, then by how many marks did he fail?
 (1) 10 (2) 15
 (3) 20 (4) 12
38. A bag contains Re 1, Rs 2, and Rs 5 coins. The quantity ratios are Re 1:Rs 2 = 2:3 and Rs 2:Rs 5 = 4:5. If the total number of coins is under 100, what is the possible total value in rupees?
 (1) 35 (2) 107
 (3) 117 (4) 70
39. 860% of 50 + 50% of 860 = ?
 (1) 430 (2) 516
 (3) 860 (4) 960

40. The price of an article is increased by 30% and 20%. It is then decreased by 10%. If its final value is Rs. 280.80. Find its initial value.

- (1) Rs. 180 (2) Rs. 220
(3) Rs. 200.57 (4) Rs. 200

41. 45% of 750 – 25% of 480 = ?

- (1) 216 (2) 217.50
(3) 236.50 (4) 245

42. If $X = 37.5\%$ of 20% of 48 and $Y = 14.28\%$ of 27.27% of 77, then

- (1) $X > Y$ (2) $X = Y$
(3) $X < Y$ (4) $X - Y = 1.4$

43. 40% of 1640 + ? = 35% of 980 + 150% of 850

- (1) 372 (2) 842
(3) 962 (4) 1052

44. When the price of rice increased by 10%, its consumption decreased by 8%. Find the percentage change on the expenditure on rice.

- (1) 2.0% increase (2) 1.0% increase
(3) 1.2% decrease (4) 1.2% increase

45. 218% of 1674 = ? $\times$ 1800

- (1) 0.5 (2) 4
(3) 6 (4) None of these

46. The ratio of the sum of the incomes of P and Q to that of Q and R is 3 : 4. The ratio of the sum of the incomes of Q and R to that of P and R is 2 : 3. If R's income is Rs. 14000, then find P's income.

- (1) Rs. 10000 (2) Rs. 8000
(3) Rs. 2000 (4) Rs. 5000

47. 60% of 264 is the same as –

- (1) 10% of 44 (2) 15% of 1056
(3) 30% of 132 (4) None of these

48. 270 candidates appeared for an competitive examination, of which 252 passed. The pass percentage is –

- (1) 80% (2) $83\frac{1}{2}\%$
(3) $90\frac{1}{3}\%$ (4) $93\frac{1}{3}\%$

49. If 90% of A = 30% of B and B = 2x% of A, then the value of x is

- (1) 450 (2) 400
(3) 300 (4) 150

50. 5 out of 2250 parts of earth is sulphur. What is the percentage of sulphur in earth?

- (1) 11/50 (2) 2/9
(3) 1/45 (4) 2/45

51. If speed increases by 33.33%, what is the percent reduction in time taken to travel the same distance?

- (1) 33.33% reduction (2) 20% reduction
(3) 50% reduction (4) 25% reduction

52. What percent of 7.2 kg is 18 gms?

- (1) 0.025% (2) 0.25%
(3) 2.5% (4) 25%

53. Consider the following equations.

$$Ax + by + c = 0$$

$$bx + cy + a = 0$$

$$cx + ay + b = 0$$

If $a + b + c \neq 0$, the given equations will be consistent for all values of a, b, c when

- (1) $ab + bc + ca = abc$ (2) $a + b + c = abc$
(3) $a^2 + b^2 + c^2 = abc$ (4) $a^2 + b^2 + c^2 = ab + bc + ca$

54. 0.01 percent is what percent of 0.1?

- (1) 1/100 (2) 1/10
(3) 10 (4) 100

55. If the area of the circle is increased by 21%, then what is the percentage increased in the circumference of the circle?

- (1) 21% (2) 10.5%
(3) 12.1% (4) 10%

56. What percent of Rs. 2650 is Rs. 1987.50?

- (1) 60% (2) 75%
(3) 80% (4) 90%

57. Ratio of prices of two houses A and B was 4 : 5 last year. This year price of A increased by 25% and that of B by Rs 50,000. If prices are now in the ratio of 9 : 10, then price of A last year was

- (1) Rs. 3,60,000 (2) Rs. 4,50,000
(3) Rs. 4,80,000 (4) Rs. 5,00,000

58. What percent of a day is 3 hours?

- (1) $12\frac{1}{2}\%$ (2) $16\frac{2}{3}\%$
(3) $18\frac{2}{3}\%$ (4) $22\frac{1}{2}\%$

59. Since prices of mangoes decreased by 25%, I can purchase 4 mangoes more for Rs. 60. What is the original price of one mango?

- (1) Rs. 4 (2) Rs. 7.5
(3) Rs. 6 (4) Rs. 5

60. It costs Re.1 to photocopy a sheet of paper. However, 2% discount is allowed on all photocopies done after first 1000 sheets. How much will it cost to copy 5000 sheets of paper?

- (1) Rs. 3920 (2) Rs. 3980
(3) Rs. 4900 (4) Rs. 4920

61. If 30% of A is added to 40% of B, the answer is 80% of B. What percentage of A is B?

- (1) 30% (2) 40%
(3) 70% (4) 75%

62. A housewife saved Rs. 2.50 in buying an item on sale. If she spent Rs. 25 for the item, approximately how much percent she saved in the transaction?

- (1) 8% (2) 9%
(3) 10% (4) 11%

63. If prices decrease by 25%, by what percentage can consumption increase without affecting the expenditure?
 (1) 25% (2) 20%
 (3) 30% (4) 33.33%
64. How many litre of pure acid are there in litre of a 20% solution?
 (1) 1.4 (2) 1.5
 (3) 1.6 (4) 2.4
65. There are 11 boxes in a row. The average number of balls per box is 30. Three of the boxes have the maximum number of balls which is 40. One of the boxes has 15 balls. The maximum number of boxes which can have no balls is
 (1) 5 (2) 4
 (3) 3 (4) 2
66. Rajeev buys goods worth Rs. 6650. He gets a rebate of 6% on it. After getting the rebate, he pays sales tax @ 10%. Find the amount he will have to pay for the goods.
 (1) Rs. 6876.10 (2) 6999.20
 (3) Rs. 6654 (4) Rs. 7000
67. If A is 37.5% more than B, by what percent is B less than A?
 (1) 37.5% (2) 30%
 (3) 25% (4) 27.27%
68. **Quantity A:** 19.9% of 85
Quantity B: 17
 (1) if quantity in column A is greater;
 (2) if quantity in column B is greater;
 (3) if quantities in both the columns are equal
 (4) if the relationship cannot be determined from the information given.
69. Vessels 1 and 2 contain milk-water mixtures with milk concentrations in the ratio 7:2. Mixing equal amounts from both yields 45% milk. In what ratio must they be mixed to achieve a 55% milk concentration?
 (1) 3 : 7 (2) 4 : 9
 (3) 7 : 3 (4) 9 : 4
70. 5% of (25% of Rs. 1600) is
 (1) Rs. 5 (2) Rs. 17.50
 (3) Rs. 20 (4) Rs. 25
71. Mira's expenditure and savings are in the ratio 3:2. Her income increases by 10%. Her expenditure increases by 12%. By what percentage does her saving increase?
 (1) 10% (2) 12%
 (3) 11% (4) 7%
72. 0.15% of $33\frac{1}{3}$ % of Rs. 10,000 is
 (1) Re. 0.05 (2) Rs. 5
 (3) Rs. 105 (4) Rs. 150
73. 17.5% of x is equal to 35
 (1) 20 (2) 50
 (3) 180 (4) 200
74. 30% of 28% of 480 is the same as
 (1) 15% of 56% of 240 (2) 60% of 28% of 240
 (3) 60% of 56% of 240 (4) None of these
75. If three-fourth of students in the class have cars and half of the car owners own a Mercedes, what fraction of the class has a Mercedes?
 (1) $\frac{1}{4}$ (2) $\frac{1}{2}$
 (3) $\frac{5}{8}$ (4) $\frac{3}{8}$
76. What is 25% of 25% equal to
 (1) 0.00625 (2) 0.0625
 (3) 0.625 (4) 6.25
77. A 40-ounce jar of milk is processed as follows: 1 ounce is drunk and replaced with water, then 2 ounces, then 3, and so on until the jar is emptied. How much total water was drunk?
 (1) 760 ounces (2) 780 ounces
 (3) 740 ounces (4) 800 ounces
78. What percent is 3% of 5%?
 (1) 15% (2) 30%
 (3) 50% (4) 60%
79. If the milk to water ratio in a mixture is 2 : 3. What is the milk percentage in the mixture?
 (1) 60% (2) 30%
 (3) 50% (4) 40%
80. 4598 is 95% of
 (1) 4800 (2) 4840
 (3) 4850 (4) 4880
81. A candidate scoring 39% fails by 58 marks. Another scoring 55% gets 22 marks above the passing threshold. Find the maximum marks of the examination.
 (1) 450 (2) 650
 (3) 500 (4) 550
82. ? % of 360 = 129.6
 (1) 36 (2) 64
 (3) 72 (4) 77
83. Amar used to spend 72% of his monthly income each month. His monthly income increased by 30% and his monthly expenditure increased by 25%. Find the percentage increase in his monthly savings.
 (1) 25% (2) 30%
 (3) 42.86% (4) 122.22%
84. ? % of $932 + 30 = 309.6$
 (1) 25 (2) 30
 (3) 35 (4) 40
85. 15% of x is equal to 15
 (1) 10 (2) 15
 (3) 85 (4) 100

86. 45% of 1500 + 35% of 1700 = ?% of 3175
 (1) 30 (2) 35
 (3) 45 (4) None of these
87. In a market survey, 20% opted for product B. The remaining individuals were uncertain. If the difference between those who opted for product B and those who were uncertain was 720, then how many individuals were covered in the survey?
 (1) 900 (2) 960
 (3) 1100 (4) 1200
88. 65% of ? = 20% of 422.50
 (1) 84.5 (2) 130
 (3) 139.425 (4) 200
89. The average of 24 numbers is 40. A new average is found after including some numbers from the set (47, 48, ..., 69). The minimum number of numbers to be included in order that the new average may be more than 45 is
 (1) 5 (2) 3
 (3) 6 (4) 7
90. An agent gets a commission of 2.5% on the sales of cloth. If on a certain day, he gets Rs. 12.50 as commission, the cloth sold through him on that day is worth
 (1) Rs. 250 (2) Rs. 500
 (3) Rs. 750 (4) Rs. 1250
91. In an election there are only two contestants A and (2) A secured 58% of the total votes and won by 8000 votes. If there were no invalid votes, what is the number of votes polled in favour of B?
 (1) 8,000 (2) 29,000
 (3) 50,000 (4) 21,000
92. If Rs. 2800 is $\frac{2}{7}$ percent of the value of a house, the worth of the house (in Rs.) is –
 (1) 8,00,000 (2) 9,80,000
 (3) 10,00,000 (4) 12,00,000
93. 8% of x is equal to 90.
 (1) 720 (2) 112.5
 (3) 1080 (4) 1125
94. 15% of ?% of 582 = 17.46
 (1) 2 (2) 10
 (3) 20 (4) None of these
95. Of the total amount received by Kiran, 20% was spending on purchases and 5% of the remaining on transportation. If he is left with Rs. 1520, the initial amount was.
 (1) 1600 (2) 1900
 (3) 2025 (4) 2000
96. The present salary per day of a worker is Rs. 264. It has increased by 10% and 20% successively in last two years. What was his salary per day before two years?
 (1) Rs. 180 (2) Rs. 220
 (3) Rs. 200 (4) None of these
97. A certain amount of money compounded annually at r% becomes Rs. 1,440 and Rs. 1,728 after 2 and 3 years respectively. Find the value of r.
 (1) 5% (2) 10%
 (3) 15% (4) 20%
98. The population of a city first increases by 25 % in one year and immediately in the next year it decreases by 20%. What was the change eventually in the population after two years?
 (1) 20 % Increase (2) 25 % Decrease
 (3) No change (4) None of these
99. The monthly salaries of Mohan and Noel were Rs. 20,000 and Rs. 25,000 respectively. Their salaries were increased by 20% and 15% respectively. Find the difference of their monthly salary after the increase.
 (1) 3750 (2) 4000
 (3) 5000 (4) 4750
100. There are several marbles in a box. If 20 % of them are red and 25 % of the rest are green then what percent of the marbles in the box are green?
 (1) 25 % (2) 33.33 %
 (3) 15 % (4) 20 %
101. Find the approximate value of the following fractions. $\frac{3}{7}$
 (1) 42.84 (2) 43.76
 (3) 44.78 (4) 41.76
102. **Quantity A:** x% of y% of z
Quantity B: y% of z% of x
 (1) if quantity in column A is greater;
 (2) if quantity in column B is greater;
 (3) if quantities in both the columns are equal
 (4) if the relationship cannot be determined from the information given.
103. In an examination consisting of 4 subjects, the marks obtained by Shyam in 3 of them are 90%, 95% and 95% respectively. Each subject is of equal marks. Under the given circumstances his average percentage marks for the examination cannot be
 (1) 94% (2) 90%
 (3) 93% (4) 96%